

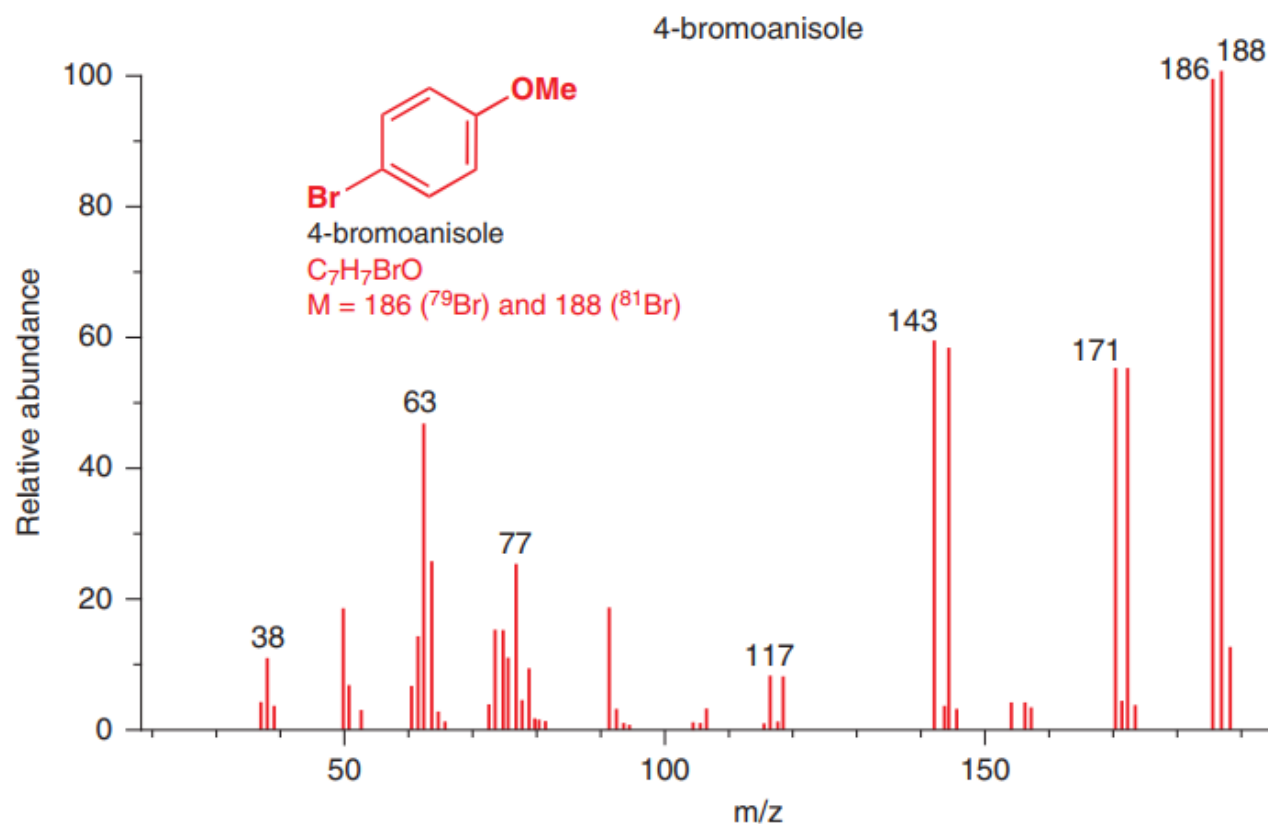
Mass Spectrometry Separates Isotopes

NATURAL ABUNDANCES OF COMMON ELEMENTS AND THEIR ISOTOPES

Element		Relative Abundance			
Hydrogen	^1H	100	^2H	0.016	
Carbon	^{12}C	100	^{13}C	1.08	
Nitrogen	^{14}N	100	^{15}N	0.38	
Oxygen	^{16}O	100	^{17}O	0.04	^{18}O 0.20
Fluorine	^{19}F	100			
Silicon	^{28}Si	100	^{29}Si	5.10	^{30}Si 3.35
Phosphorus	^{31}P	100			
Sulfur	^{32}S	100	^{33}S	0.78	^{34}S 4.40
Chlorine	^{35}Cl	100			^{37}Cl 32.5
Bromine	^{79}Br	100			^{81}Br 98.0
Iodine	^{127}I	100			

Mass spectrometry detects isotopes

Most elements can exist as more than one isotope. Usually, one isotope accounts for the vast majority (perhaps >99%) of the atoms of an element. But for some elements, atoms of several isotopes make up a significant proportion of the total in a sample. Chlorine, for example, is normally a 3:1 mixture of ^{35}Cl and ^{37}Cl , while bromine is an almost 1:1 mixture of ^{79}Br and ^{81}Br . Because mass spectrometry weighs individual molecules, there is no averaging: instead it detects the true weight of each molecule, whatever isotope it contains.



El mass spectrum

two chlorine atoms

1. The Decimal Format (0.75×0.25)

This shows the probability of each specific pairing occurring in the molecule:

- **Peak M** ($^{35}\text{Cl} + ^{35}\text{Cl}$):

$$0.75 \times 0.75 = \mathbf{0.5625}$$

- **Peak M+2 (Mixed isotopes):**

There are two ways to get this (^{35}Cl then ^{37}Cl , OR ^{37}Cl then ^{35}Cl):

$$(0.75 \times 0.25) + (0.25 \times 0.75) = 0.1875 + 0.1875 = \mathbf{0.3750}$$

- **Peak M+4** ($^{37}\text{Cl} + ^{37}\text{Cl}$):

$$0.25 \times 0.25 = \mathbf{0.0625}$$

2. The Ratio Format (3×1)

Since ^{35}Cl is 3 times more abundant than ^{37}Cl , we can use whole numbers (3 and 1) to find the relative ratio:

- **Peak M:** $3 \times 3 = \mathbf{9}$
- **Peak M+2:** $(3 \times 1) + (1 \times 3) = \mathbf{6}$
- **Peak M+4:** $1 \times 1 = \mathbf{1}$

Resulting Ratio: 9 : 6 : 1

three chlorine atoms

1. The Decimal Format (0.75×0.25)

This calculates the probability of picking specific isotopes three times in a row:

- **Peak M ($3 \times {}^{35}\text{Cl}$):**
 $0.75 \times 0.75 \times 0.75 = \mathbf{0.421875}$
- **Peak M+2 ($2 \times {}^{35}\text{Cl}$ and $1 \times {}^{37}\text{Cl}$):**
There are 3 ways to arrange this (35-35-37, 35-37-35, 37-35-35).
 $3 \times (0.75 \times 0.75 \times 0.25) = \mathbf{0.421875}$
- **Peak M+4 ($1 \times {}^{35}\text{Cl}$ and $2 \times {}^{37}\text{Cl}$):**
There are 3 ways to arrange this (37-37-35, 37-35-37, 35-37-37).
 $3 \times (0.75 \times 0.25 \times 0.25) = \mathbf{0.140625}$
- **Peak M+6 ($3 \times {}^{37}\text{Cl}$):**
 $0.25 \times 0.25 \times 0.25 = \mathbf{0.015625}$

2. The Ratio Format (3×1)

Using the 3 : 1 abundance ratio makes the math much cleaner to see as a set of whole numbers:

- **Peak M:** $3 \times 3 \times 3 = \mathbf{27}$
- **Peak M+2:** $3 \times (3 \times 3 \times 1) = \mathbf{27}$
- **Peak M+4:** $3 \times (3 \times 1 \times 1) = \mathbf{9}$
- **Peak M+6:** $1 \times 1 \times 1 = \mathbf{1}$

The resulting ratio is **27 : 27 : 9 : 1**.

For a compound containing **two Bromine atoms**

Peak	Isotope Combination	Statistical Probability	Relative Intensity
M	$^{79}\text{Br} + ^{79}\text{Br}$	$0.5 \times 0.5 = 0.25$	1
M+2	$(^{79}\text{Br} + ^{81}\text{Br}) + (^{81}\text{Br} + ^{79}\text{Br})$	$(0.5 \times 0.5) + (0.5 \times 0.5) = 0.50$	2
M+4	$^{81}\text{Br} + ^{81}\text{Br}$	$0.5 \times 0.5 = 0.25$	1

- **Peak M :**

$$1 \text{ (from } ^{79}\text{Br)} \times 1 \text{ (from } ^{79}\text{Br)} = \mathbf{1}$$

- **Peak $M + 2$:**

$$(1 \times 1) + (1 \times 1) = 1 + 1 = \mathbf{2}$$

- **Peak $M + 4$:**

$$1 \text{ (from } ^{81}\text{Br)} \times 1 \text{ (from } ^{81}\text{Br)} = \mathbf{1}$$

Resulting Ratio: 1 : 2 : 1

For a compound containing **three Bromine atoms**

1. The Decimal Format ($0.5 \times 0.5 \times 0.5$)

This shows the probability of each combination occurring in nature:

- **Peak M** ($3 \times {}^{79}\text{Br}$): $0.5 \times 0.5 \times 0.5 = \mathbf{0.125}$
- **Peak M+2** ($2 \times {}^{79}\text{Br}$ and $1 \times {}^{81}\text{Br}$):
There are 3 combinations (79-79-81, 79-81-79, 81-79-79). $3 \times (0.5 \times 0.5 \times 0.5) = \mathbf{0.375}$
- **Peak M+4** ($1 \times {}^{79}\text{Br}$ and $2 \times {}^{81}\text{Br}$):
There are 3 combinations (81-81-79, 81-79-81, 79-81-81). $3 \times (0.5 \times 0.5 \times 0.5) = \mathbf{0.375}$
- **Peak M+6** ($3 \times {}^{81}\text{Br}$): $0.5 \times 0.5 \times 0.5 = \mathbf{0.125}$

2. The Ratio Format ($1 \times 1 \times 1$)

Since the isotopes are in a **1:1** ratio, we can use simple units:

- **Peak M:** $1 \times 1 \times 1 = \mathbf{1}$
- **Peak M+2:** $(1 \times 1 \times 1) \times 3 \text{ ways} = \mathbf{3}$
- **Peak M+4:** $(1 \times 1 \times 1) \times 3 \text{ ways} = \mathbf{3}$
- **Peak M+6:** $1 \times 1 \times 1 = \mathbf{1}$

Resulting Ratio: 1 : 3 : 3 : 1

one Bromine and one Chlorine (BrCl)

1. The Decimal Format (0.5×0.75)

We use the abundance of Bromine ($0.5 : 0.5$) and Chlorine ($0.75 : 0.25$):

- **Peak M** ($^{79}\text{Br} + ^{35}\text{Cl}$):

$$0.5 \times 0.75 = \mathbf{0.375}$$

- **Peak M+2 (Two different combinations):**

- $^{81}\text{Br} + ^{35}\text{Cl} \rightarrow 0.5 \times 0.75 = 0.375$

- $^{79}\text{Br} + ^{37}\text{Cl} \rightarrow 0.5 \times 0.25 = 0.125$

- **Total for M+2:** $0.375 + 0.125 = \mathbf{0.500}$

- **Peak M+4** ($^{81}\text{Br} + ^{37}\text{Cl}$):

$$0.5 \times 0.25 = \mathbf{0.125}$$

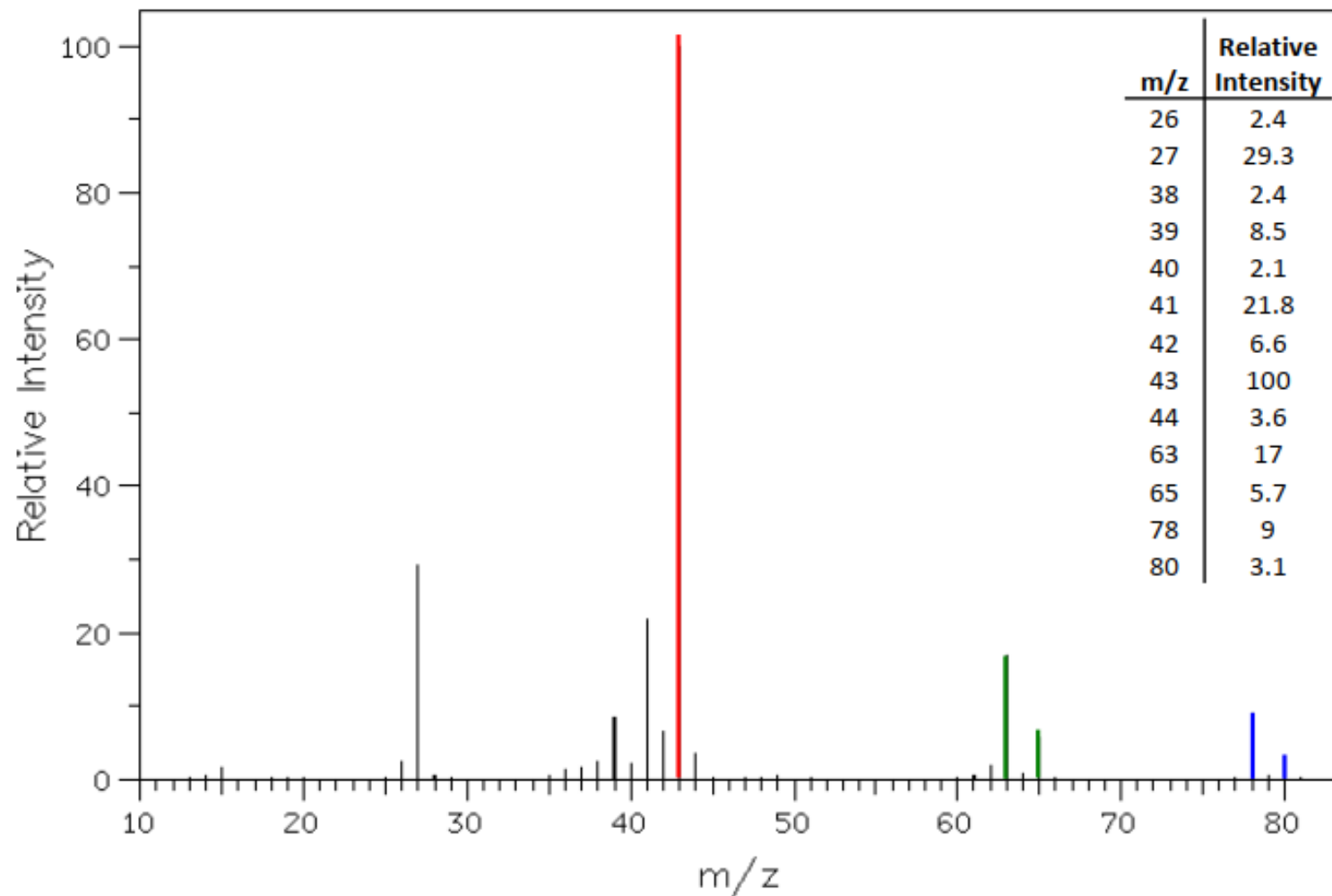
2. The Ratio Format (1×3)

Using the relative abundance ratios ($1 : 1$ for Br and $3 : 1$ for Cl) gives us clean whole numbers:

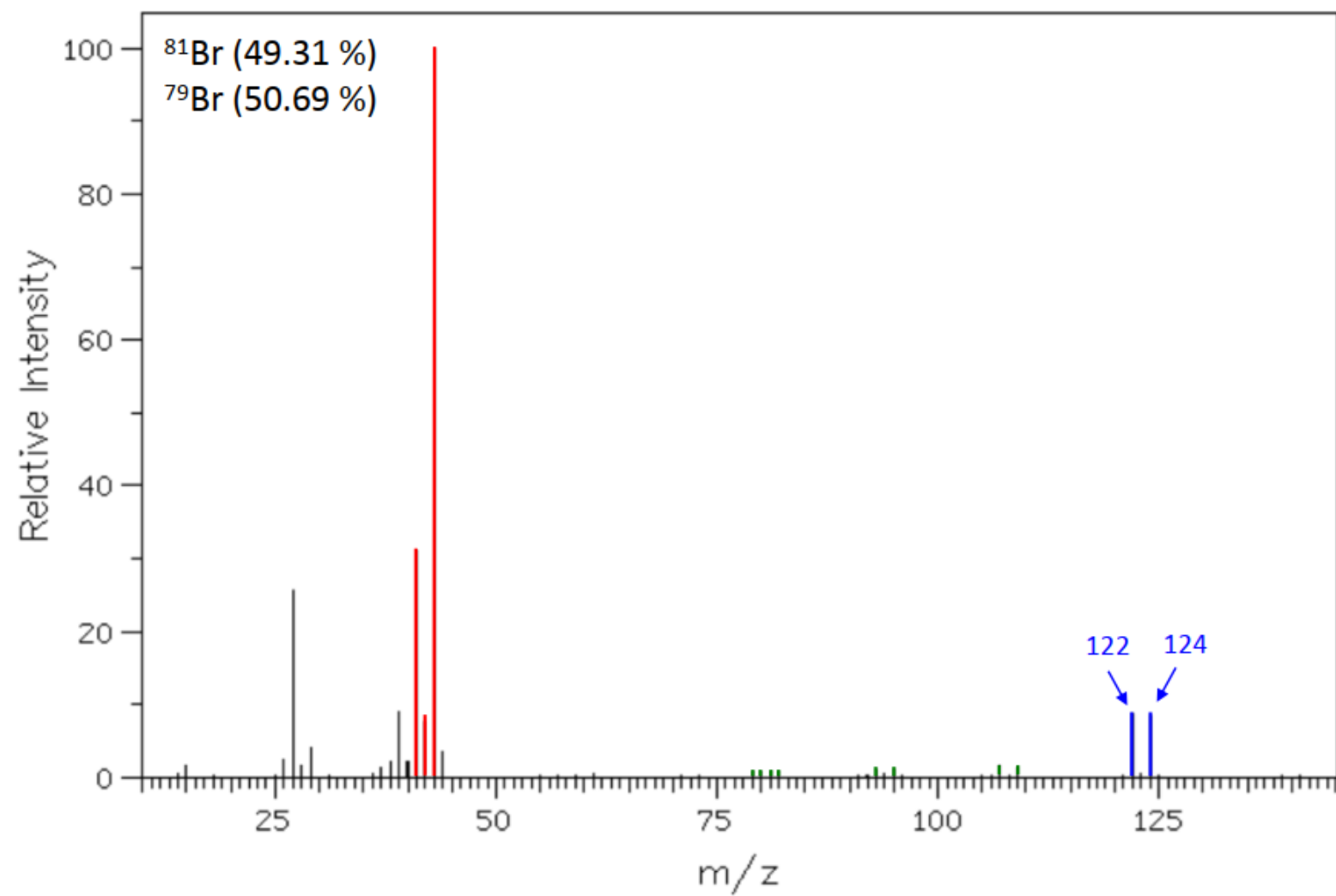
- **Peak M:** $1 \times 3 = \mathbf{3}$
- **Peak M+2:** $(1 \times 3) + (1 \times 1) = 3 + 1 = \mathbf{4}$
- **Peak M+4:** $1 \times 1 = \mathbf{1}$

The resulting ratio is **3 : 4 : 1**.

The mass spectrum of a chlorine-containing molecule is likewise easy to identify from two peaks separated by two mass units, but this time in a ratio of 3:1, arising from the 3:1 isotopic ratio of ^{35}Cl and ^{37}Cl .



EI-Mass Spectrum of Isopropyl Chloride



EI-Mass Spectrum of Propyl Bromide



Q.7	An organic compound exhibits the $[M]^+$, $[M+2]^+$ and $[M+4]^+$ peaks in the intensity ratio 1:2:1 in the mass spectrum, and shows a singlet at δ 7.49 in the ^1H NMR spectrum in CDCl_3 . The compound is:
(A)	1,4-dichlorobenzene
(B)	1,4-dibromobenzene
(C)	1,2-dibromobenzene
(D)	1,2-dichlorobenzene

NET June-2018

Mass spectrum of a compound shows an $[M+2]$ ion peak that is 25% of M . This indicates that compound has one—

a) Fluorine b) Sulphur c) Bromine b) Chlorine

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Mass fragment of $[\text{IrCl}]^+$ in mass spectrometry shows three mass peaks at $m/z = 226, 228$, and 230 . Given that natural abundances of ^{191}Ir , ^{193}Ir , ^{35}Cl , and ^{37}Cl are 37%, 63%, 76%, and 24% respectively, the intensities of the mass peaks are in the order

1. 49.5: 100: 26.6
2. 100: 49.5: 26.6
3. 26.6: 100: 49.5
4. 26.6: 49.5: 100

This calculates the probability of each mass peak occurring:

- **Peak M ($^{191}\text{Ir} + ^{35}\text{Cl}$):**
 $0.37 \times 0.75 = \mathbf{0.2775}$
- **Peak M+2 (Two combinations):**
 - $^{193}\text{Ir} + ^{35}\text{Cl} \rightarrow 0.63 \times 0.75 = 0.4725$
 - $^{191}\text{Ir} + ^{37}\text{Cl} \rightarrow 0.37 \times 0.25 = 0.0925$
 - **Total for M+2:** $0.4725 + 0.0925 = \mathbf{0.5650}$
- **Peak M+4 ($^{193}\text{Ir} + ^{37}\text{Cl}$):**
 $0.63 \times 0.25 = \mathbf{0.1575}$

Ratio Format (Abundance Units)

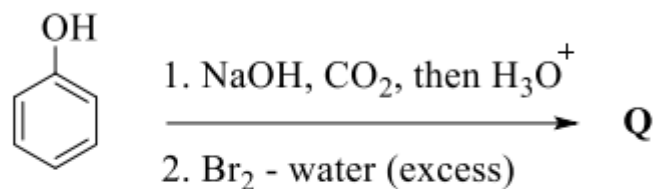
Using the approximate whole-number ratios (Ir: 37/63 vs Cl: 75/25):

- **Peak M:** $37 \times 75 = \mathbf{2775}$
- **Peak M+2:** $(63 \times 75) + (37 \times 25) = 4725 + 925 = \mathbf{5650}$
- **Peak M+4:** $63 \times 25 = \mathbf{1575}$

Q.16 The major product in the given reaction sequence is **Q**. The mass spectrum of **Q** shows

Ans: B

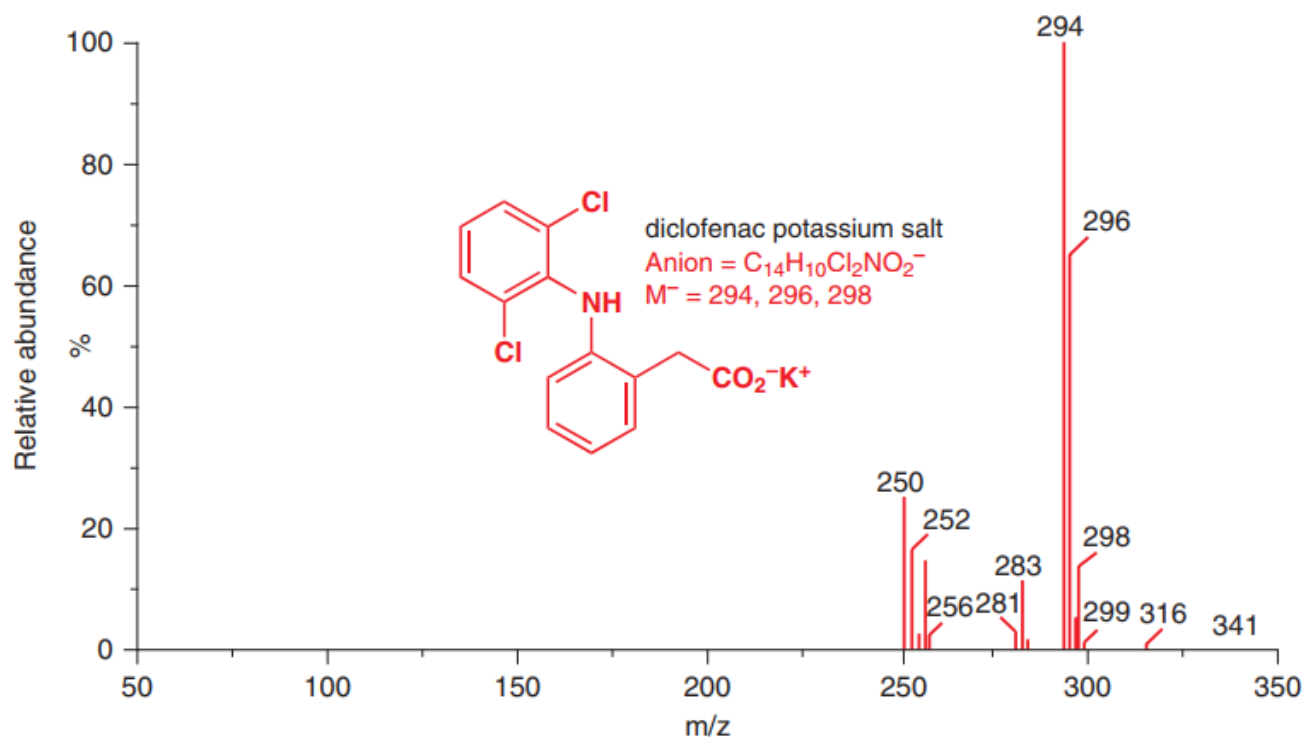
([M] = molecular ion peak)



- (A) [M], [M+2], [M+4], and [M+6] peaks with relative intensity of 1:1:1:1
- (B) [M], [M+2], [M+4], and [M+6] peaks with relative intensity of 1:3:3:1
- (C) [M], [M+2], and [M+4] peaks with relative intensity of 1:2:1
- (D) [M] and [M+2] peaks with relative intensity of 1:1

What happens with more than one Br or Cl?

The ES spectrum of diclofenac shows the mass of the carboxylate anion as three peaks, at 294, 296, and 298. The relative size of the peaks can be worked out from the 75% probability that each Cl atom will be ^{35}Cl and the 25% probability it will be ^{37}Cl . The ratios are therefore $3/4 \times 3/4$: $2 \times 3/4 \times 1/4$: $1/4 \times 1/4$ or 9 : 6 : 1.



Summary table of common elements with more than one isotope at >1% abundance

Element	Isotopes	Approximate ratio	Exact ratio
carbon	^{12}C , ^{13}C		98.9:1.1
chlorine	^{35}Cl , ^{37}Cl	3:1	75.8:24.2
bromine	^{79}Br , ^{81}Br	1:1	50.5:49.5

H, N, O, S, P, F, and I have only very small amounts of isotopes other than ^1H , ^{14}N , ^{16}O , ^{31}P , ^{32}S , and ^{127}I . The real oddity though is tin, which exists as a mixture of 10 different stable isotopes, the major ones being ^{116}Sn (15%), ^{117}Sn (8%), ^{118}Sn (24%), ^{119}Sn (9%), ^{120}Sn (33%), ^{122}Sn (5%), and ^{124}Sn (6%). In reality the precise ratio of isotopes for any element varies according to its source, a fact which can supply useful forensic information.

CSIR-DECEMBER-2015

In the mass spectrum of 1,2-dichloroethane, approximate ratio of peaks at m/z values 98, 100, 102 will be

- | | |
|----------|----------|
| 1. 3:1:1 | 2. 9:6:1 |
| 3. 1:1:2 | 4. 1:2:1 |

What is the approximate ratio of the peaks at m/z values 98, 100, and 102 in the mass spectrum of 1,2-dichloroethane?

In the mass spectrum of 1,2-dichloroethane, approximate ratio of peaks at m/z values 98, 100, 102 will be---

Fragmentation and Structural Analysis

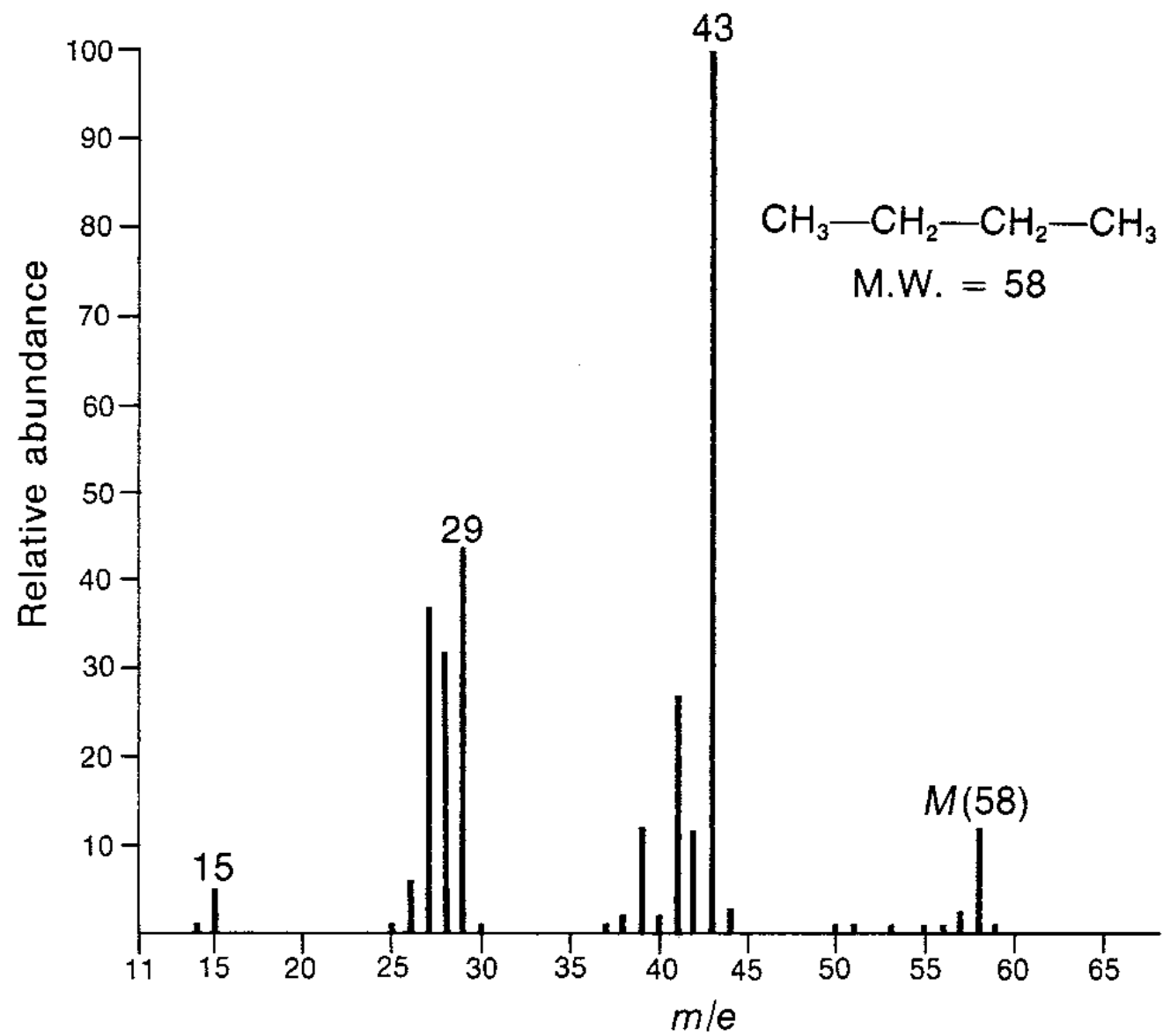
In electron ionization mass spectrometry (EI-MS):

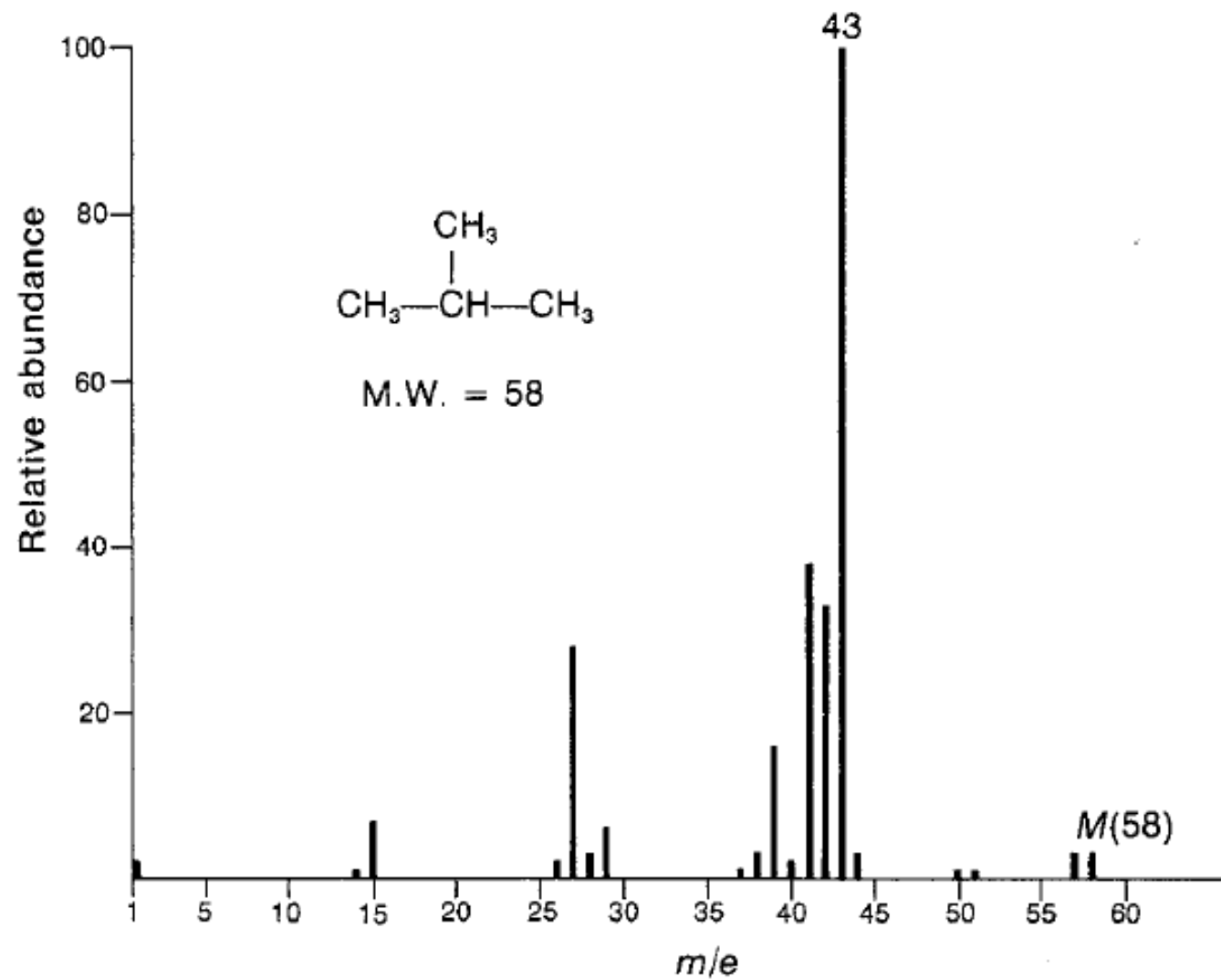
- A molecule is bombarded by high-energy electrons in the ionization chamber
- A radical cation is formed as the sample molecule loses an electron
- The molecule also absorbs an extra amount of energy
- This extra energy fragments the radical cation into smaller pieces

Fragmentation Pattern

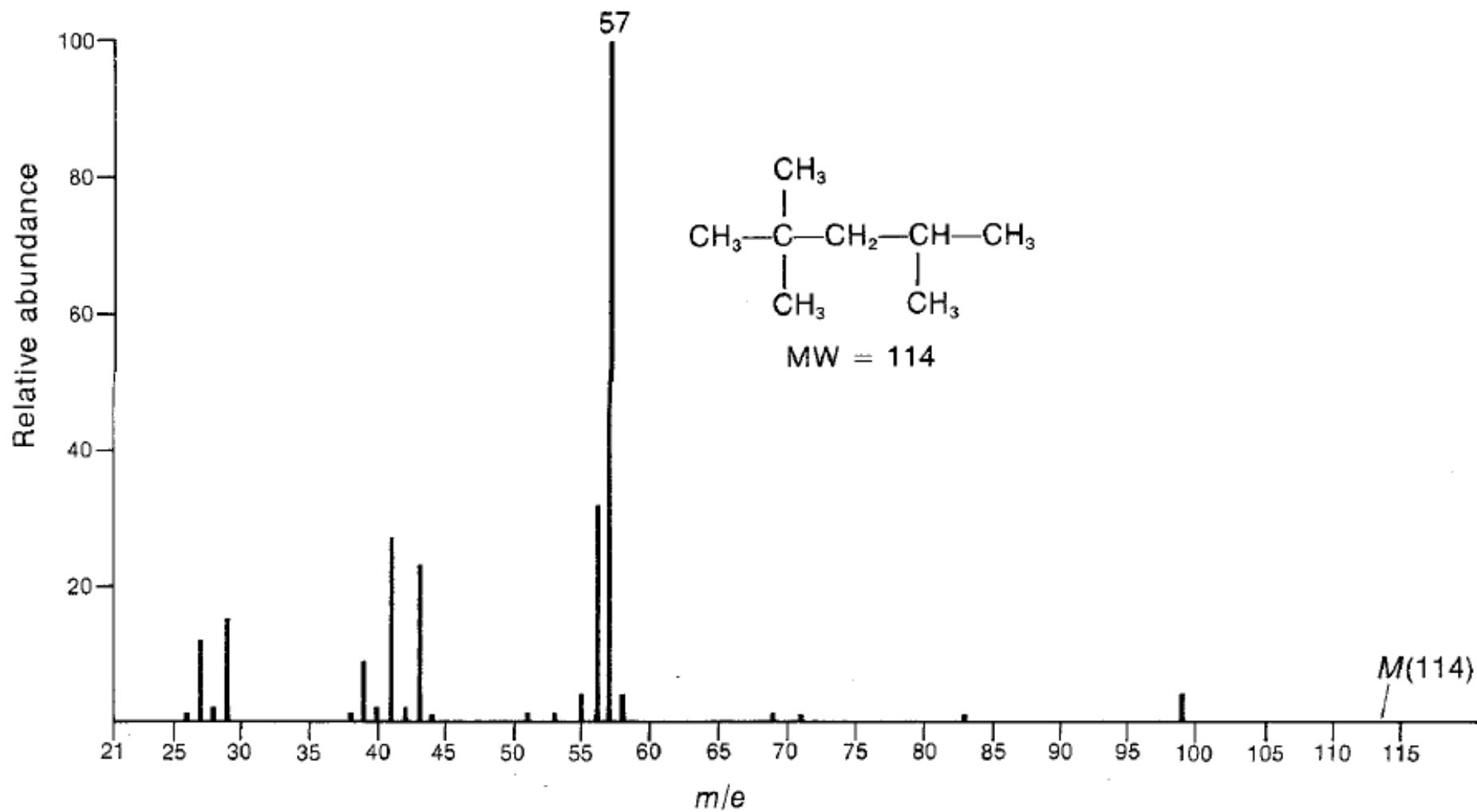


Structural Analysis

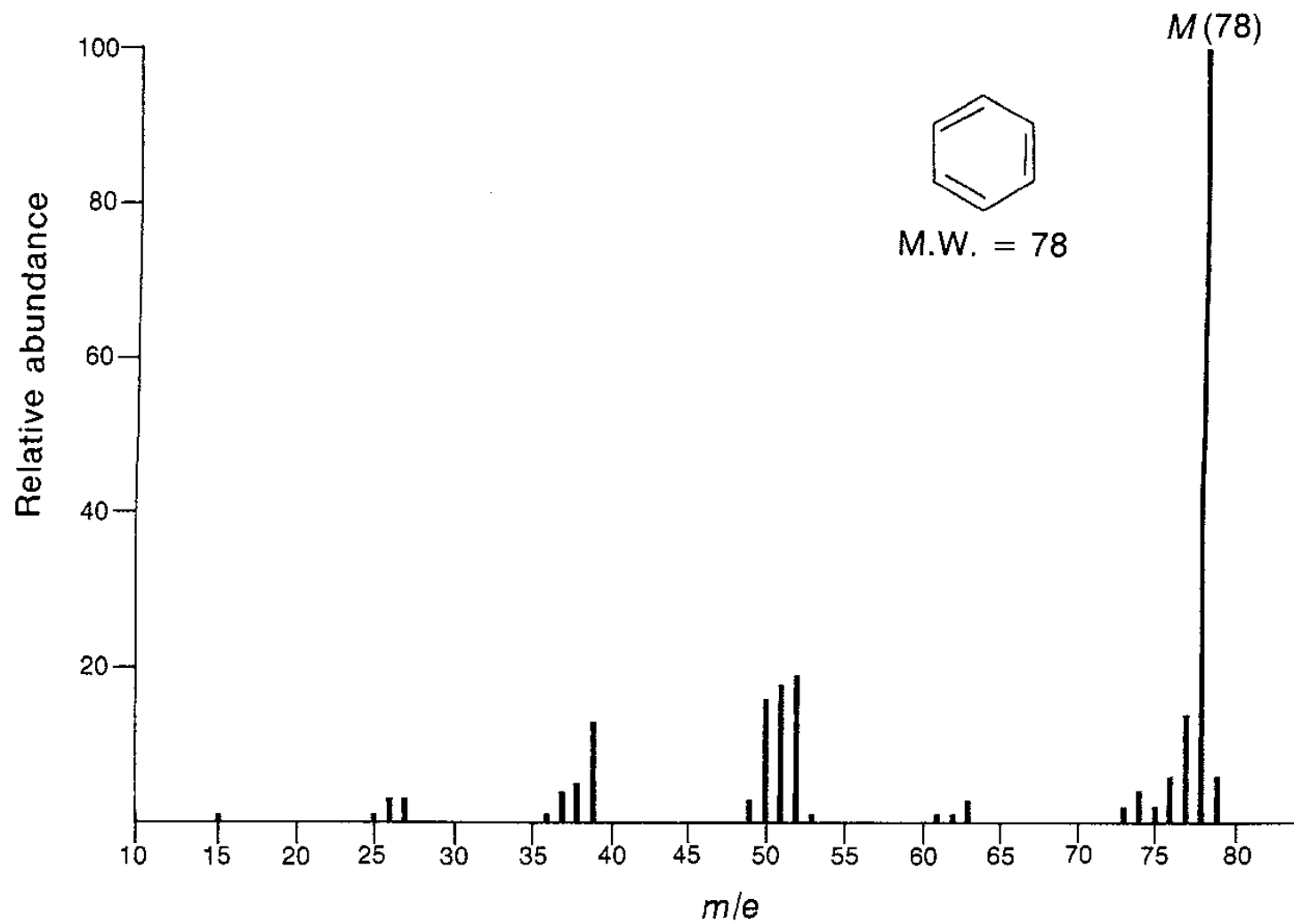




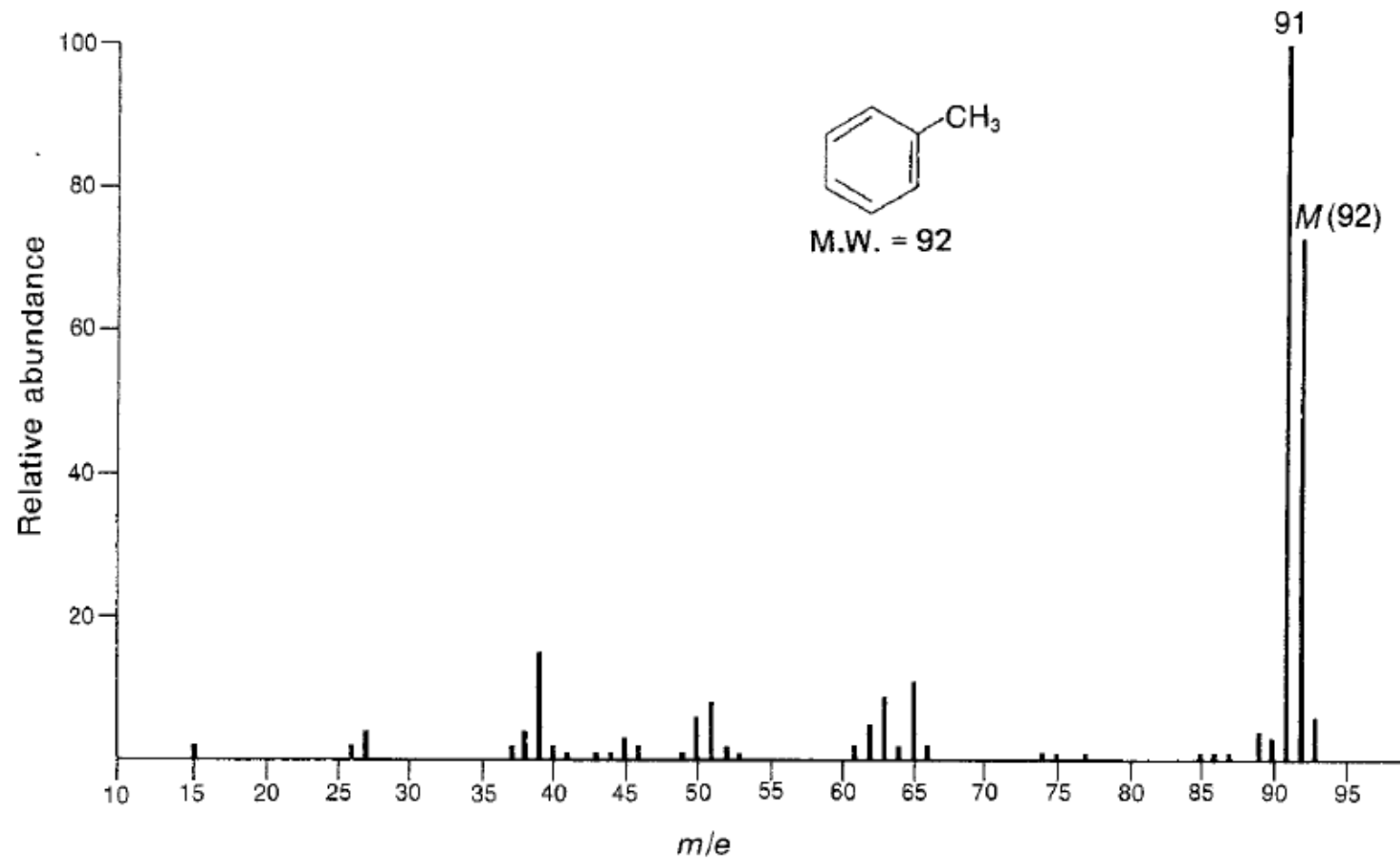
The intensity of the molecular ion peak is low



Molecular ion peak is not observed

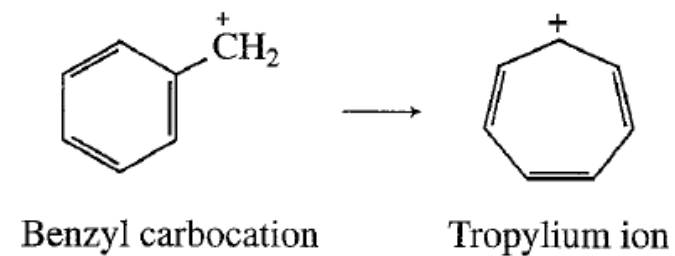


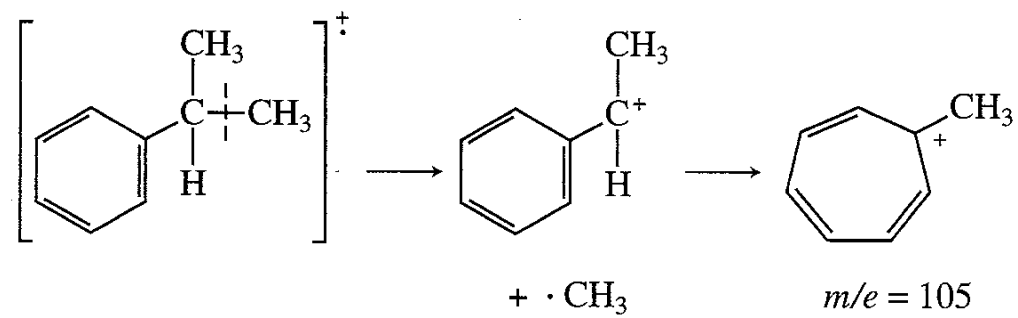
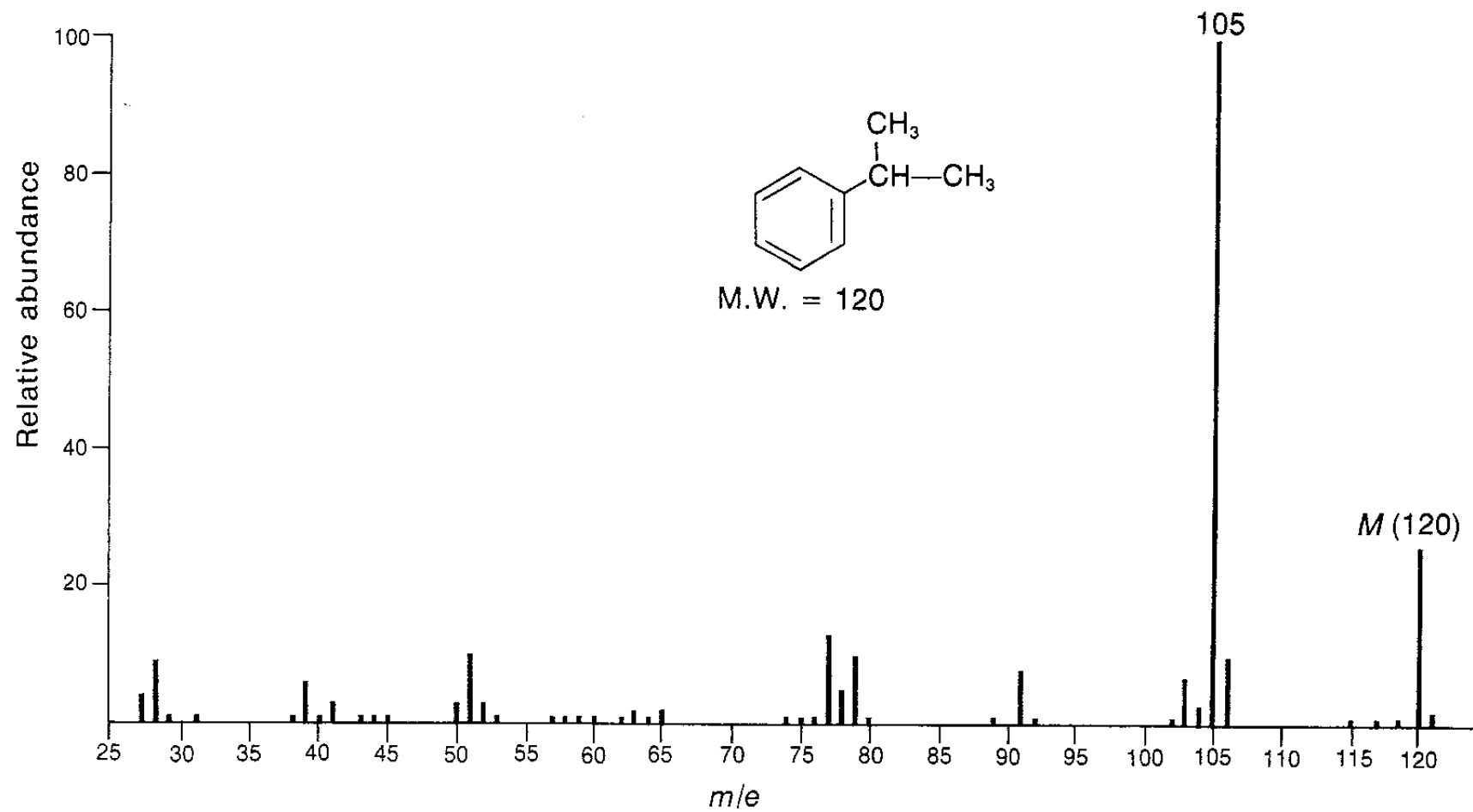
Aromatic hydrocarbons show intense molecular ion peak

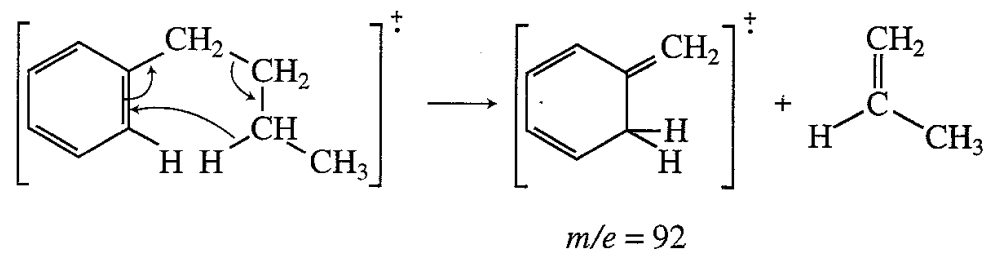
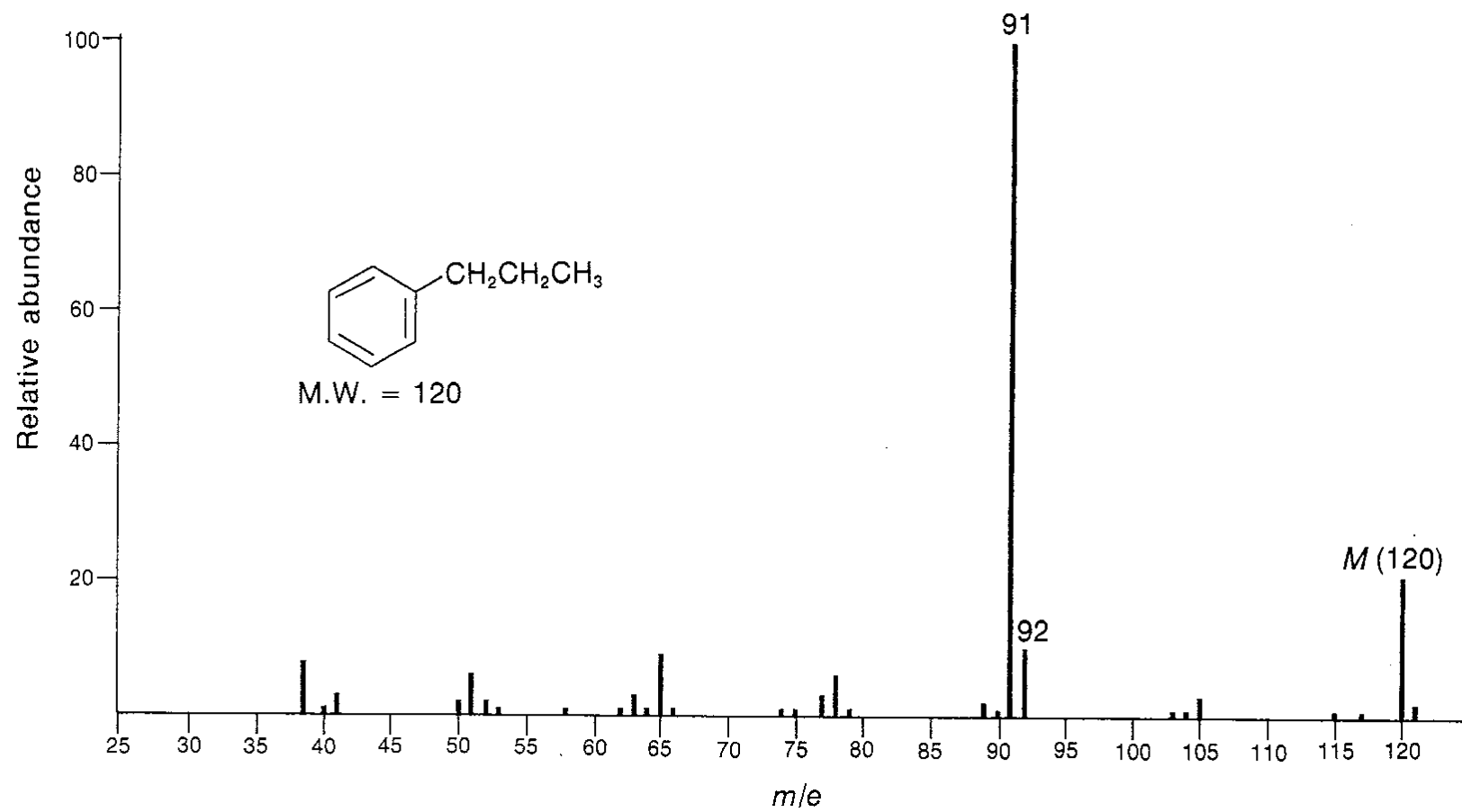


Is the peak at $M = 91$ for a benzyl cation?

It is actually not!







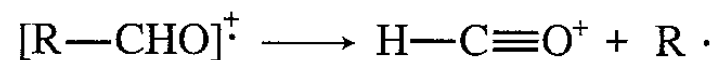
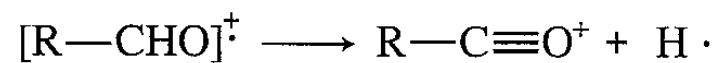
Alkyl chain-length 3 or more

McLafferty Rearrangement

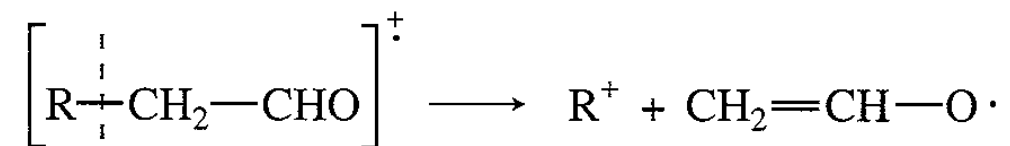
Fragmentation in Aldehydes and Ketones

Three main types of fragmentations:

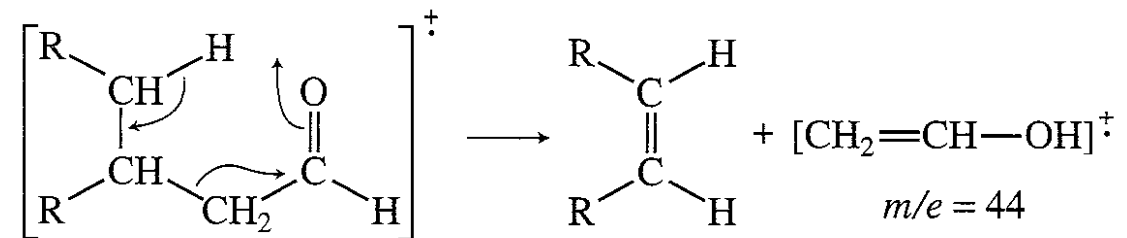
α-cleavage:

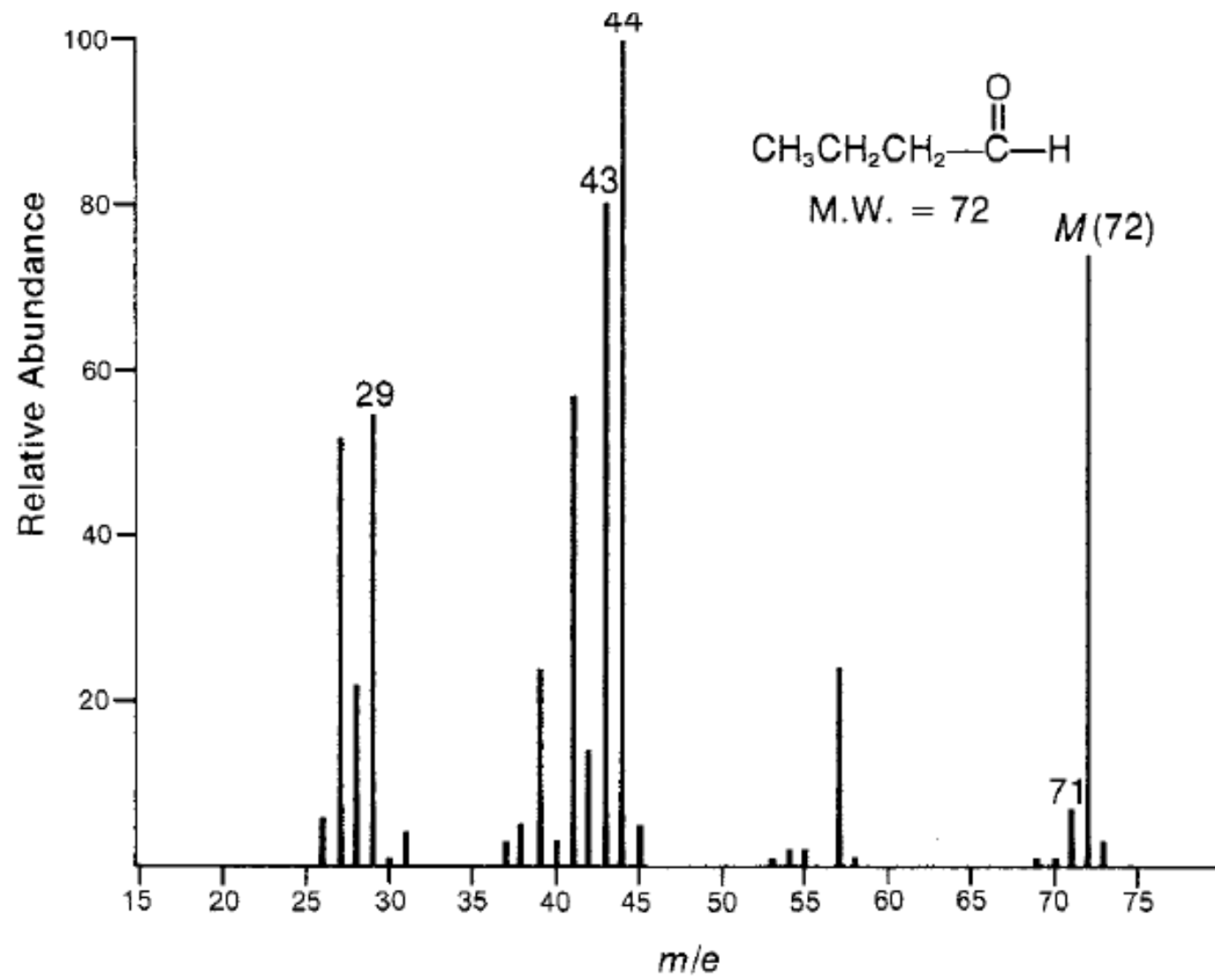


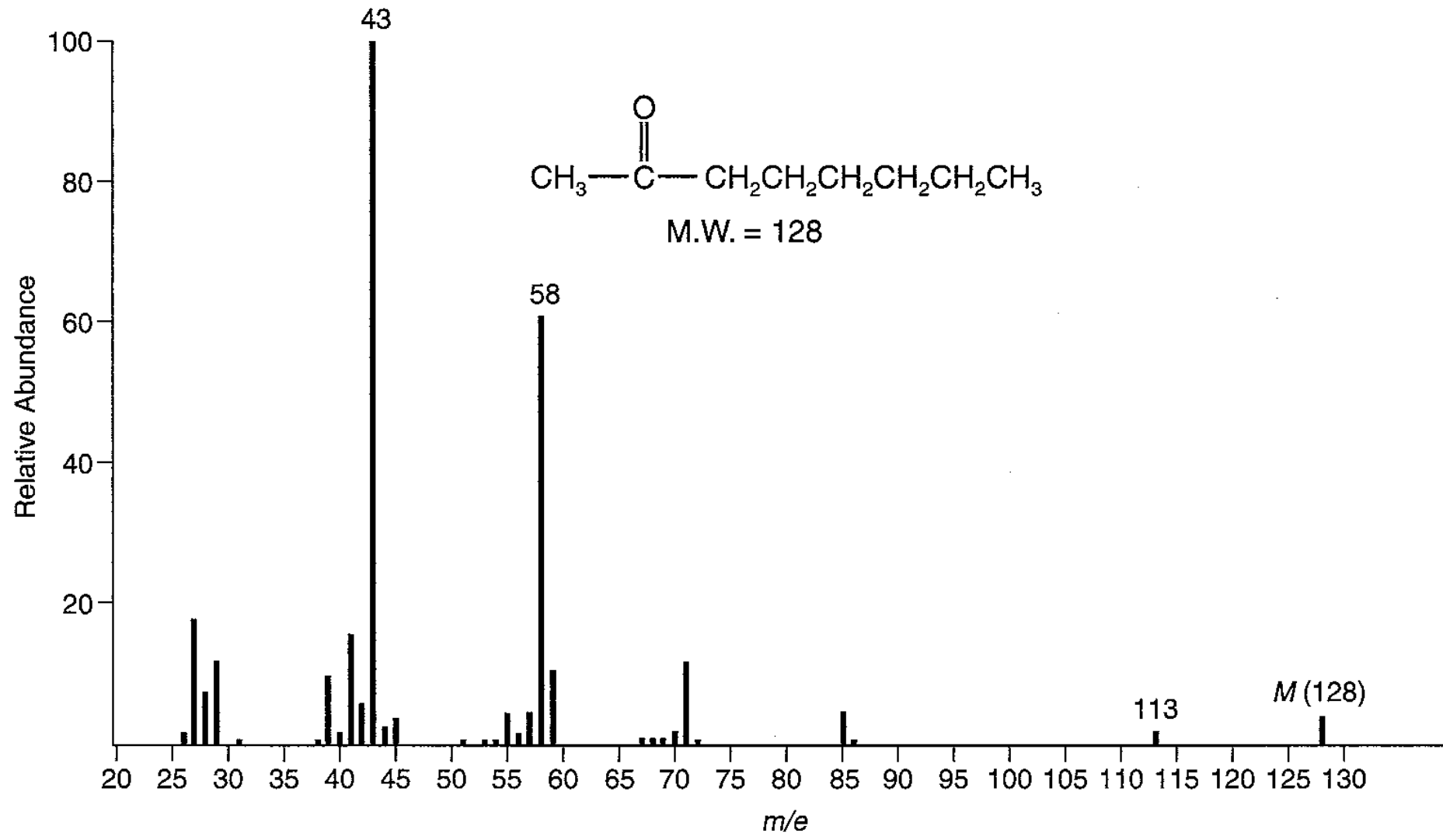
β-cleavage:



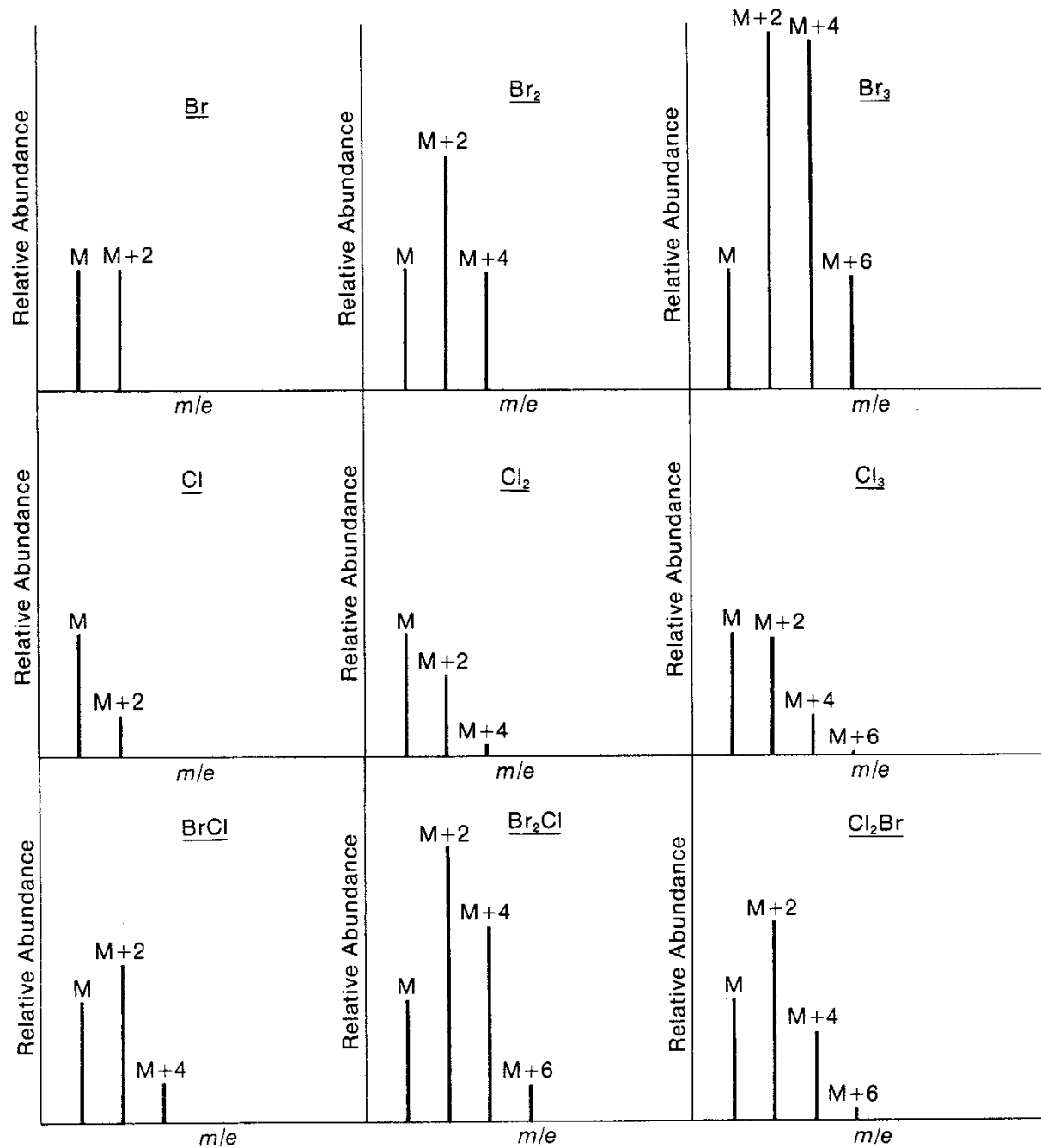
McLafferty Rearrangement







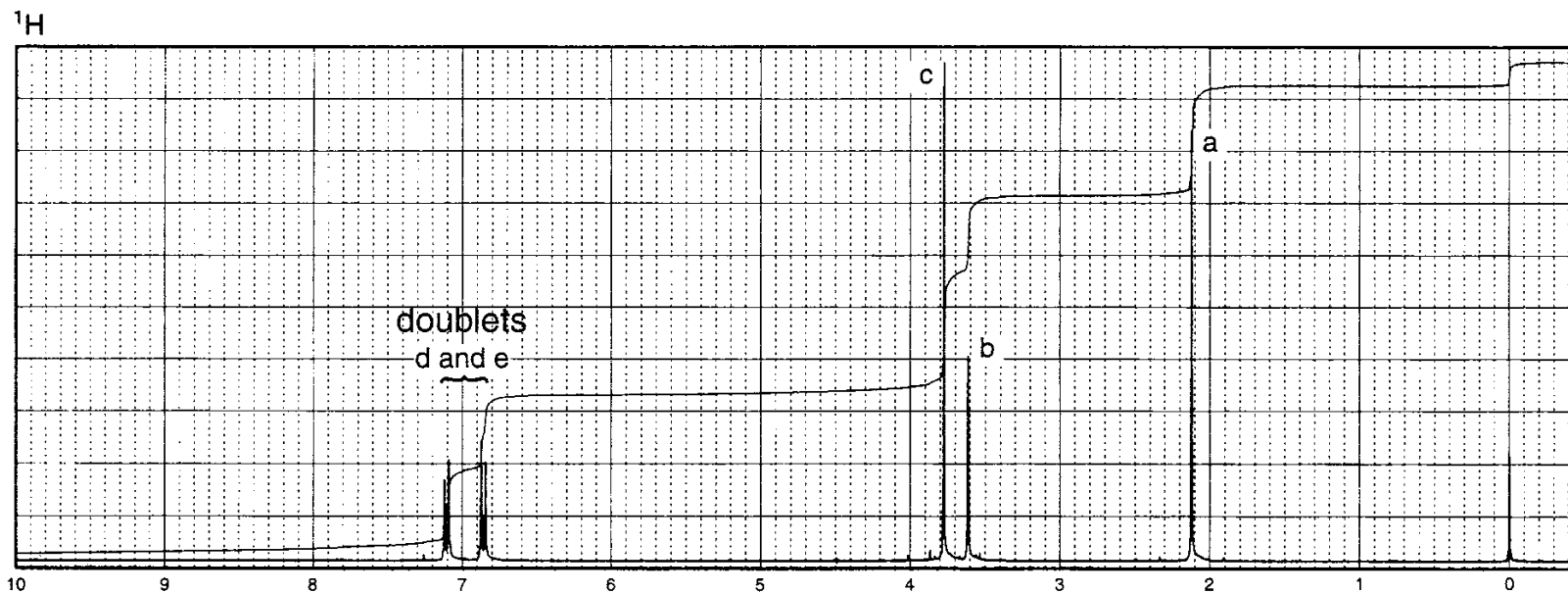
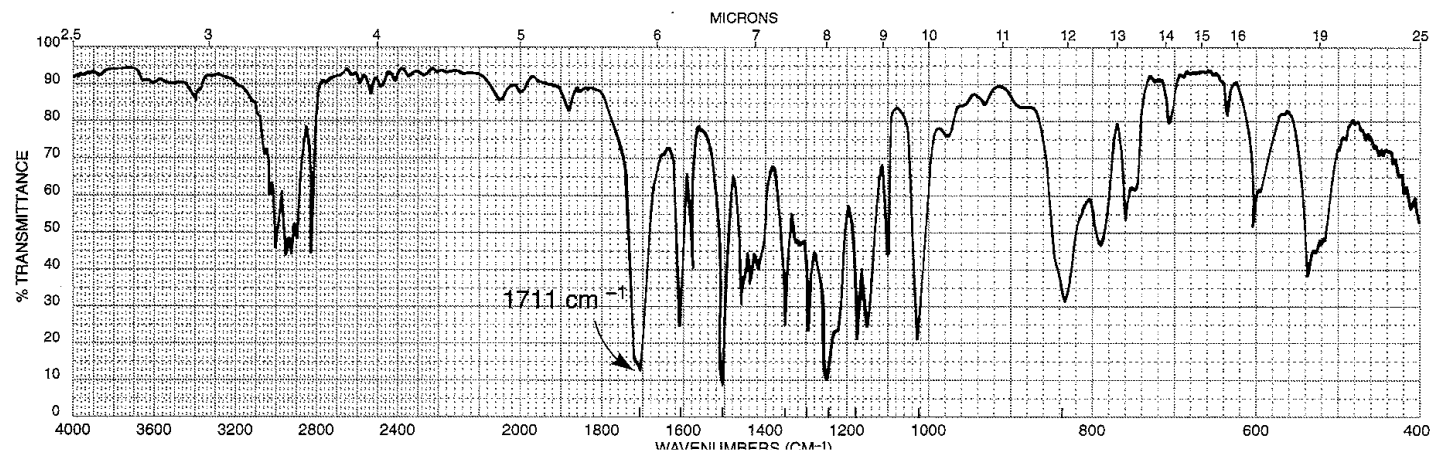
Compounds with Multiple Br/Cl Atoms



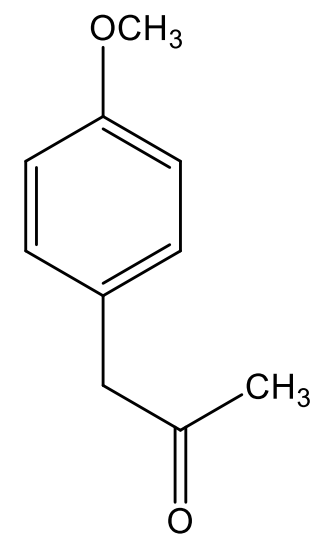
RELATIVE INTENSITIES OF ISOTOPE PEAKS FOR VARIOUS COMBINATIONS OF BROMINE AND CHLORINE

Halogen	Relative Intensities			
	M	$M + 2$	$M + 4$	$M + 6$
Br	100	97.7		
Br ₂	100	195.0	95.4	
Br ₃	100	293.0	286.0	93.4
Cl	100	32.6		
Cl ₂	100	65.3	10.6	
Cl ₃	100	97.8	31.9	3.47
BrCl	100	130.0	31.9	
Br ₂ Cl	100	228.0	159.0	31.2
Cl ₂ Br	100	163.0	74.4	10.4

Q. A compound shows a molecular ion peak M in its mass spectrum at 164. The M+1 peak has a relative intensity of 10.9%. If only C, H and O are present, what is the molecular formula of the compound? Determine the structure of the compound from the IR, proton NMR, carbon NMR and DEPT spectral data provided below. The integration ratio for the peaks (proton NMR) at 2.1, 3.6, 3.8, 6.85 and 7.1 is 3:2:3:2:2. Assign all the peaks in the proton and carbon NMR spectra.



Normal Carbon	DEPT-135	DEPT-90
29 ppm	Positive	No peak
50	Negative	No peak
55	Positive	No peak
114	Positive	Positive
126	No peak	No peak
130	Positive	Positive
159	No peak	No peak
207	No peak	No peak



Mass spectrometry

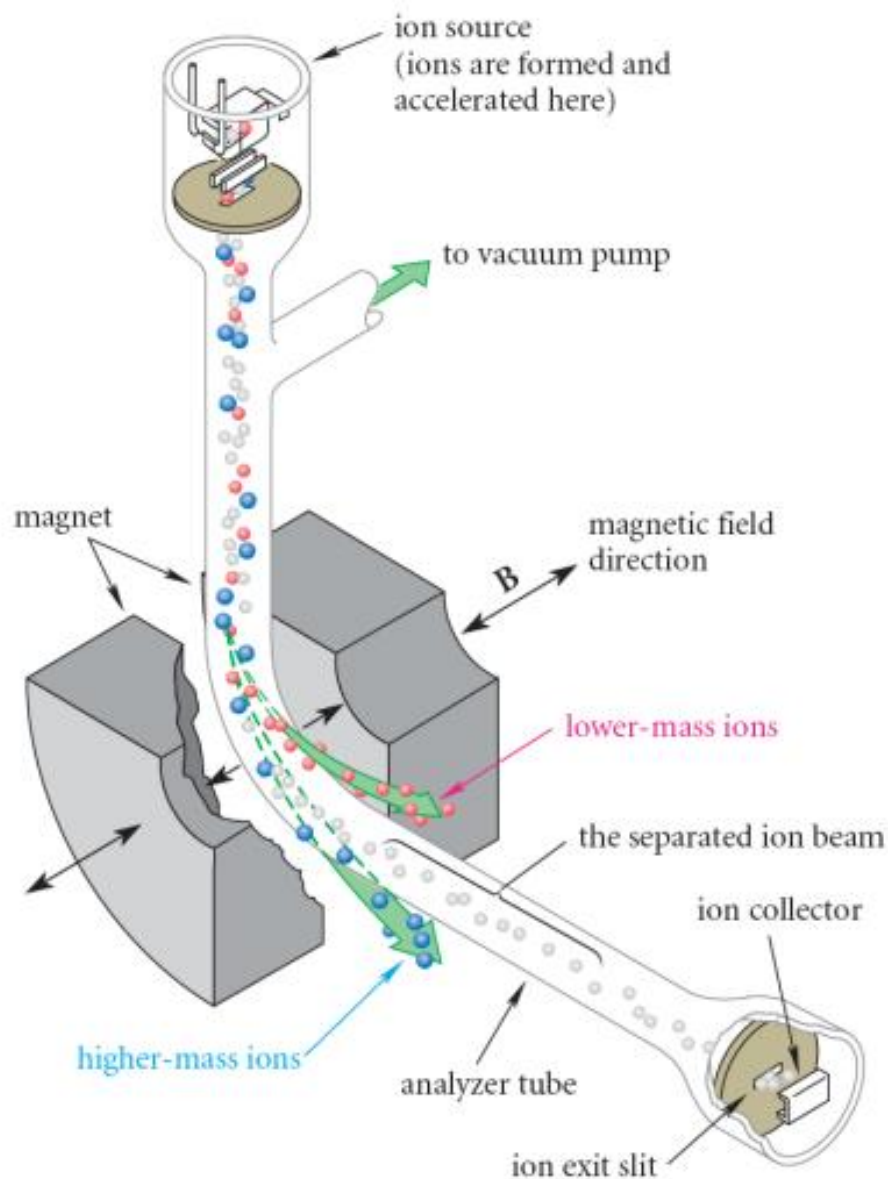
Mass spectrometry weighs the molecule

It is not easy to weigh a neutral molecule, and a mass spectrometer works by measuring the mass of a charged ion instead: the charge makes the molecule controllable by an electric field. A mass spectrometer therefore has three basic components:

- something to volatilize and ionize the molecule into a beam of charged particles
- something to focus the beam so that particles of the same mass:charge ratio are separated from all others
- something to detect the particles

All spectrometers in common use operate in a high vacuum and use one of several methods to convert neutral molecules into cations, the most common being electron impact, chemical ionization, and electrospray.

Electron Impact Mass Spectrometry (EI-MS)

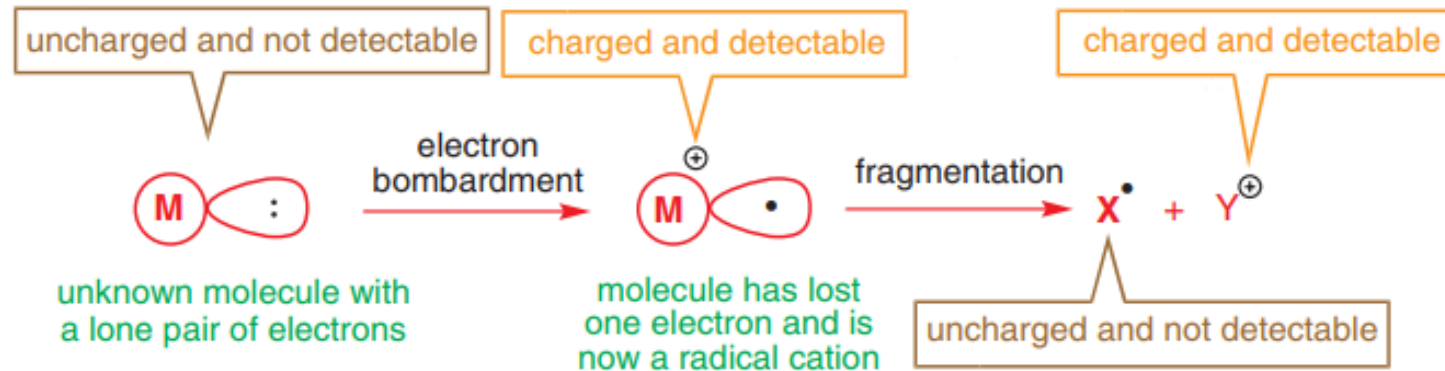


Electron-Impact Mass Spectrometry (EI-MS) is used to determine molecular masses and connectivity by bombarding a molecule with high energy electrons and detecting the masses of the resulting fragments.

- Highly sensitive and destructive.
- Ionization from a 70 eV beam of electrons.
- Ions formed are controlled by a magnetic field, mass selected, and detected at an ion collector.
- Detection does not use electromagnetic radiation (not spectroscopy).
- Fragmentation provides structural and molecular mass information.

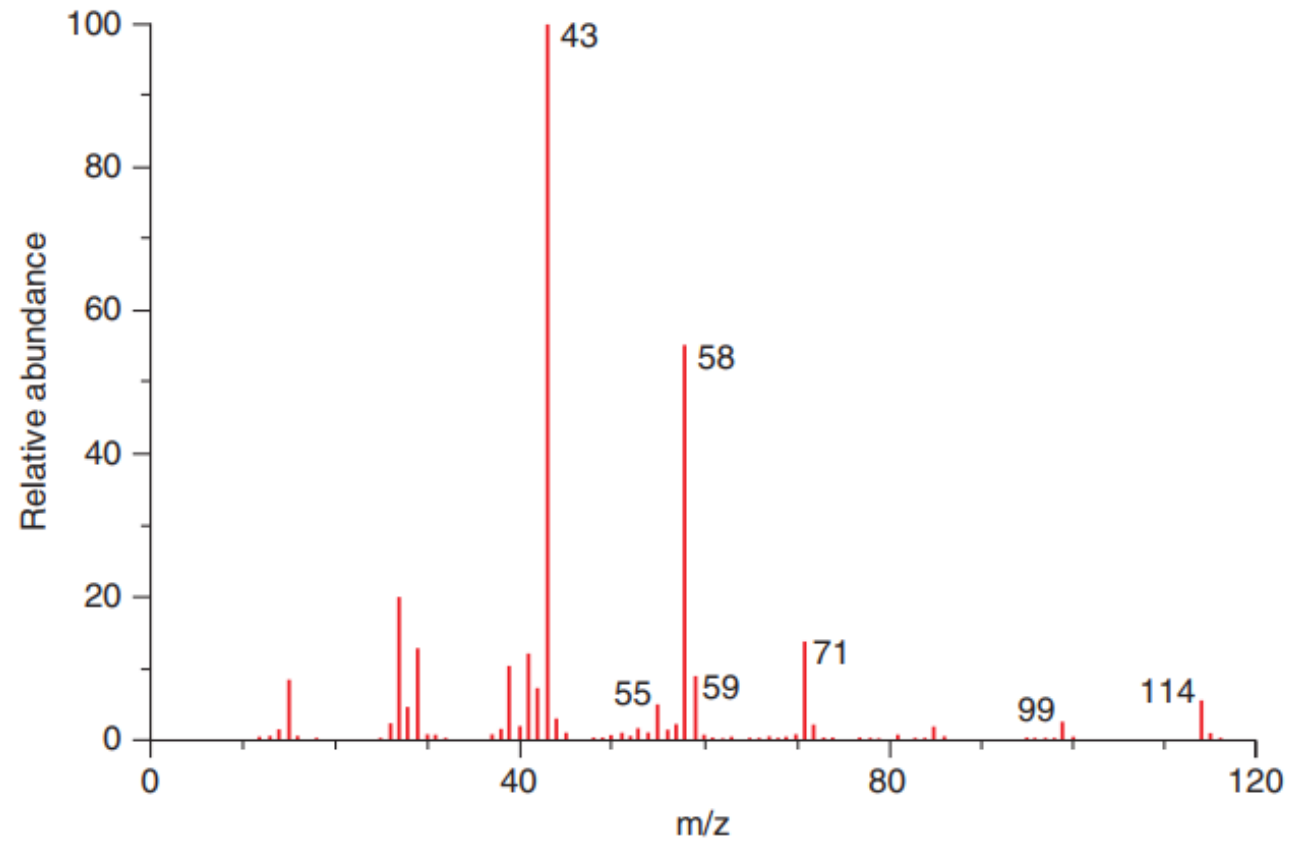
Mass spectrometry by electron impact

In electron impact (EI) mass spectrometry the molecule is bombarded with highly energetic electrons that knock a weakly bound electron out of the molecule. If you think this is strange, think of throwing bricks at a brick wall: the bricks can't stick to the wall but can knock loose bricks off the top of the wall. Losing a single electron leaves behind an unpaired electron and a positive charge. The electron that is lost will be one of relatively high energy, and typically one not involved in bonding, for example an electron from a lone pair.

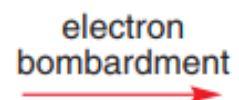


Thus ammonia gives $\text{NH}_3^{+\bullet}$ and a ketone gives $\text{R}_2\text{CO}^{+\bullet}$. These unstable species are known as radical cations, and being charged they are accelerated by an electric field and focused onto the detector, which detects the mass of the ion by how far its path has been deflected by the electric field. It only takes about 20 μs for the radical cations to reach the detector, but sometimes they fragment before they get there, in which case other ions will also be detected. These fragments will always have a lower mass than the 'parent' molecular ion, so in a typical mass spectrum we are most interested in the heaviest ion we can see.

Mass spectrum of honey bee alarm pheromone (electron impact)



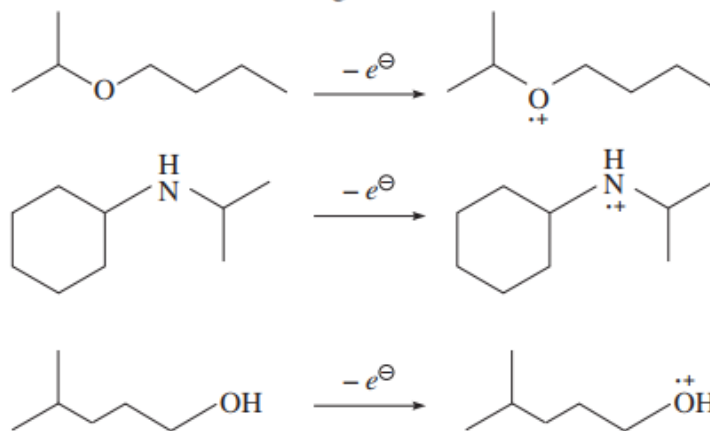
heptan-2-one



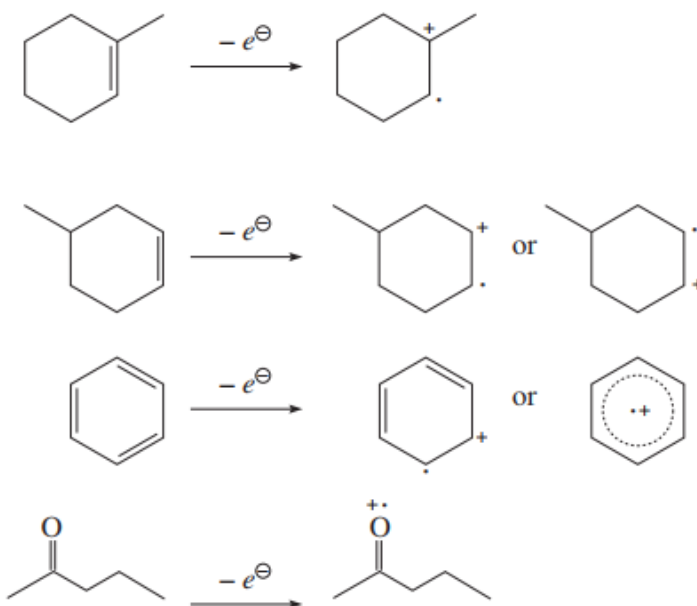
$M^{+•} = 114 = C_7H_{14}O$

The electrons most likely to be ejected during the ionization event are the ones that are in the highest potential energy molecular orbitals, that is the electrons held least tightly by the molecule. Thus, it is easier to remove an electron from a nonbonding orbital, n , than it is to strip an electron from a p orbital. Similarly it is much easier to eject an electron from a p orbital in comparison to a s orbital. The molecular ion can be represented with either a localized or a non-localized charge site. Some examples of loss of an electron and the notation for the molecular ion are shown below.

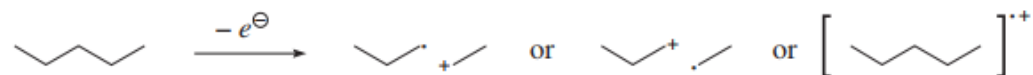
Loss of an electron from a non-bonding orbital:



Loss of an electron from a π orbital:



Loss of an electron from a σ orbital:

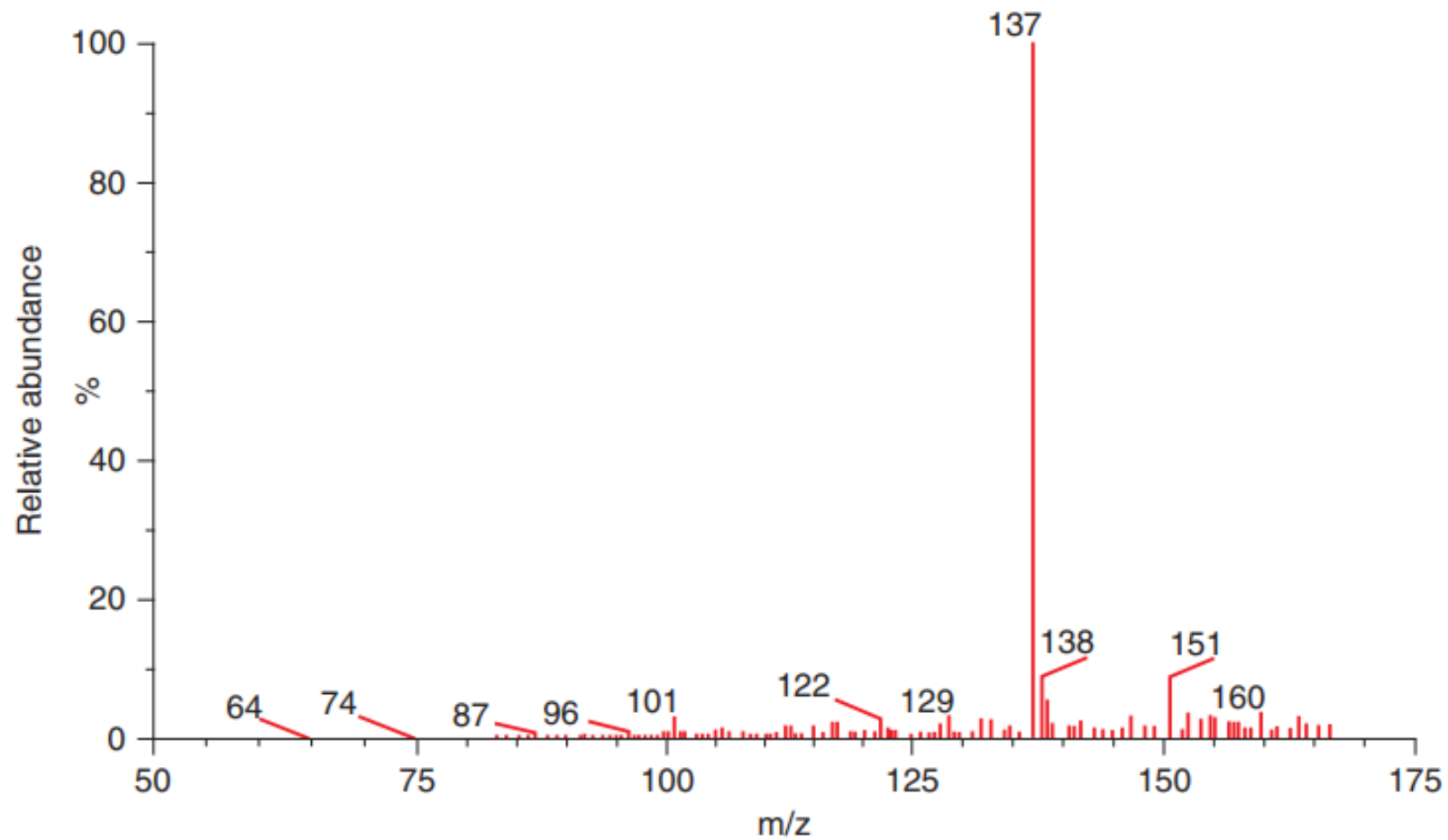


Mass spectrometry by chemical ionization, electrospray, or other methods

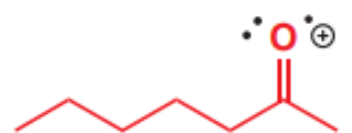
A problem with EI mass spectrometry is that, for fragile molecules, the energy of the bombarding electron can be sufficient to cause it to fragment completely, losing all trace of the molecular ion. Some useful information can be gained from fragmentation patterns, but in general it is more useful to aim to weigh the molecule all in one piece. This can be achieved using any of a number of other techniques, of which the most common are chemical ionization (CI) and electrospray (ES).

Chemical ionization is achieved by mixing a gas such as ammonia with the substrate in the spectrometer. Bombardment of NH_3 with electrons leads to formation of some NH_4^+ by proton transfer, and reaction of this ion with the substrate makes a charged complex, which can be accelerated by the electric field. The masses observed by chemical ionization spectroscopy carried out in this way are usually $M + 1$ or $M + 18$ (the mass of NH_4^+) relative to the mass of the substrate. With electrospray mass spectroscopy, an aerosol of the substrate is ionized, and ionization in the presence of sodium ions means that masses of $M + 1$ and $M + 23$ are often seen, or, if the ionization forms anions, $M - 1$.

Electrospray mass spectrum of heptan-2-one



electron
bombardment



Principle of Mass Spectrometry – Summary

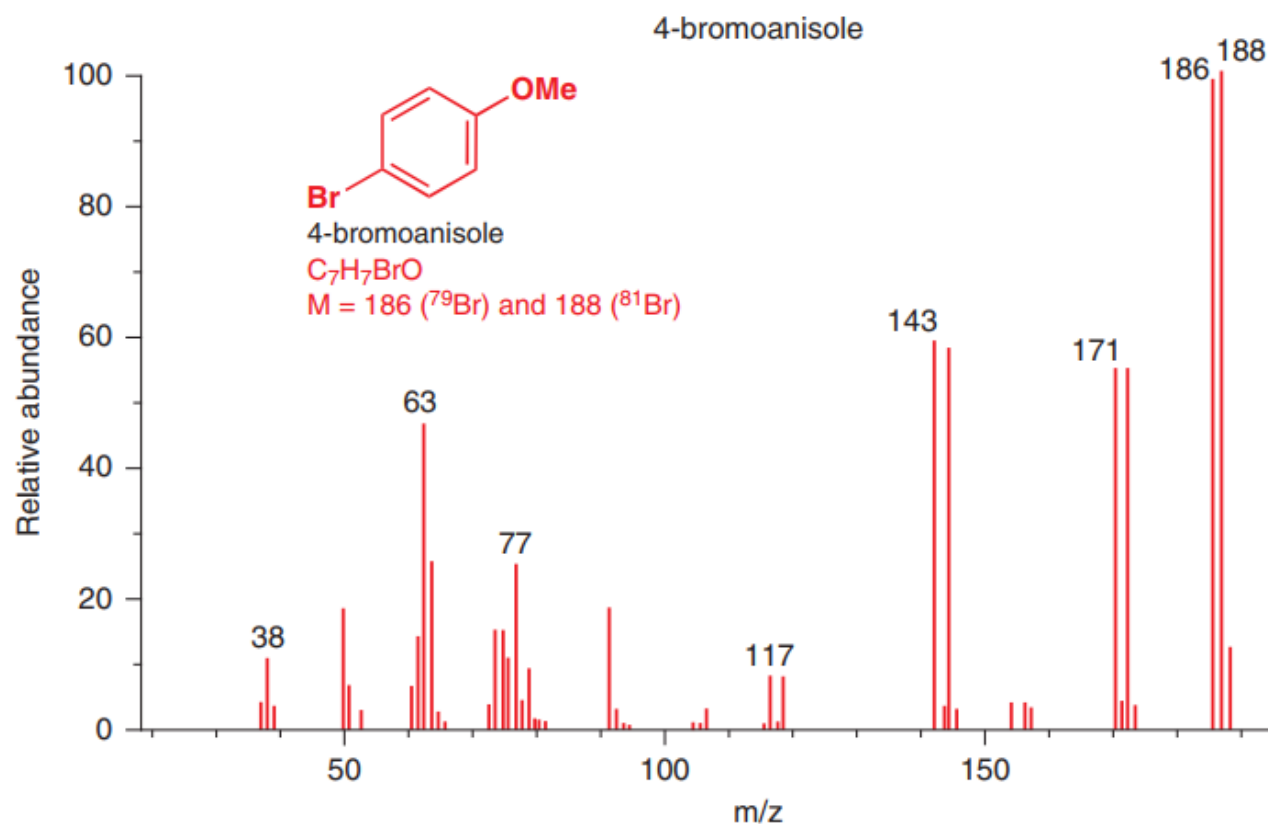
- Mass spectroscopy (MS) ionizes and energizes a molecule to determine its mass and connectivity through fragmentation.



- Electron impact- Mass Spectroscopy (EI-MS) uses high energy electrons to ionize the molecules.
- Only cationic fragments are responsive to the magnetic field and detected.
- Abundance of the ion detected is dependent upon their rate of formation and their rate of fragmentation.
- More stable ions that are easy to form tend to be responsible for more intense signals.

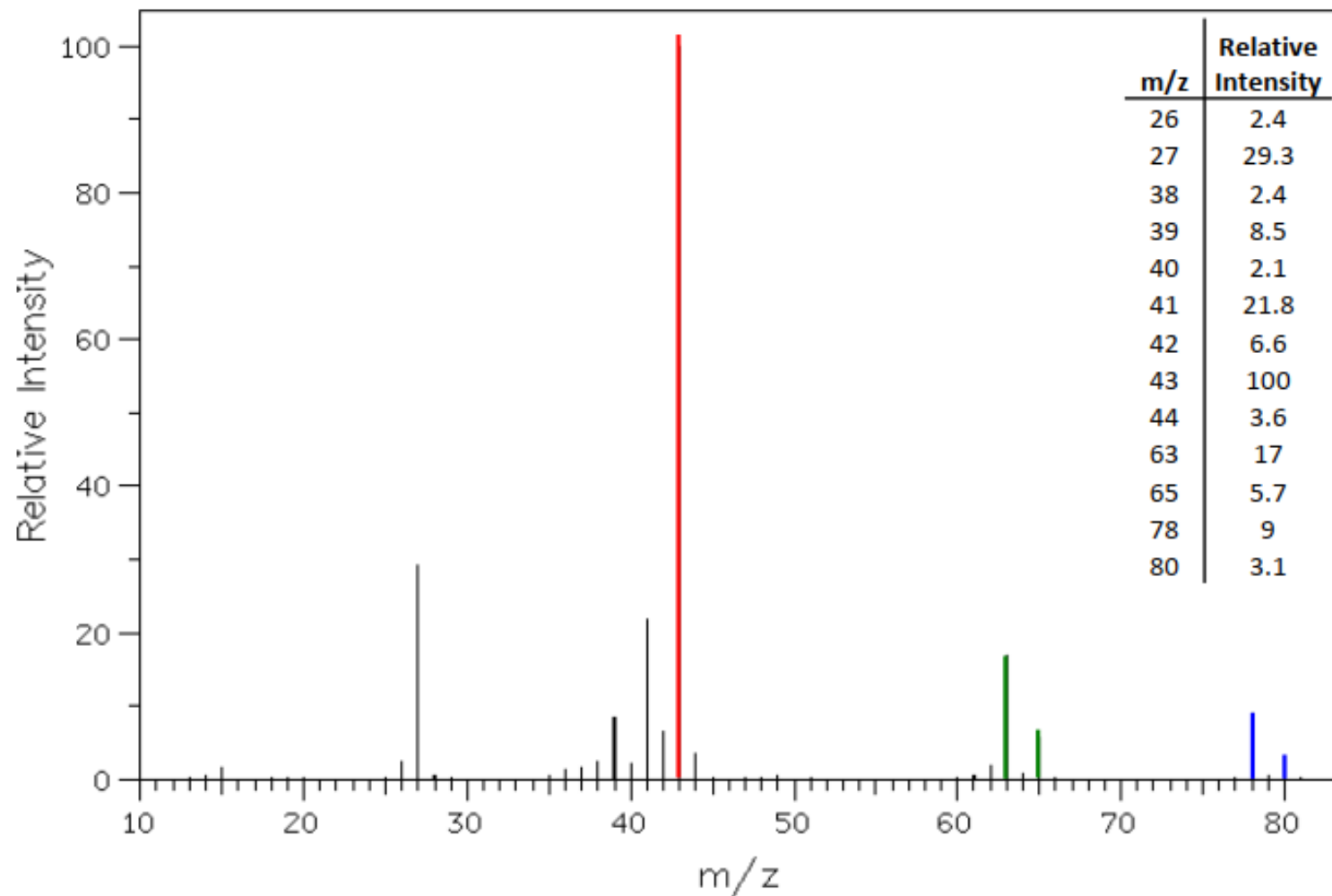
Mass spectrometry detects isotopes

Most elements can exist as more than one isotope. Usually, one isotope accounts for the vast majority (perhaps >99%) of the atoms of an element. But for some elements, atoms of several isotopes make up a significant proportion of the total in a sample. Chlorine, for example, is normally a 3:1 mixture of ^{35}Cl and ^{37}Cl , while bromine is an almost 1:1 mixture of ^{79}Br and ^{81}Br . Because mass spectrometry weighs individual molecules, there is no averaging: instead it detects the true weight of each molecule, whatever isotope it contains.

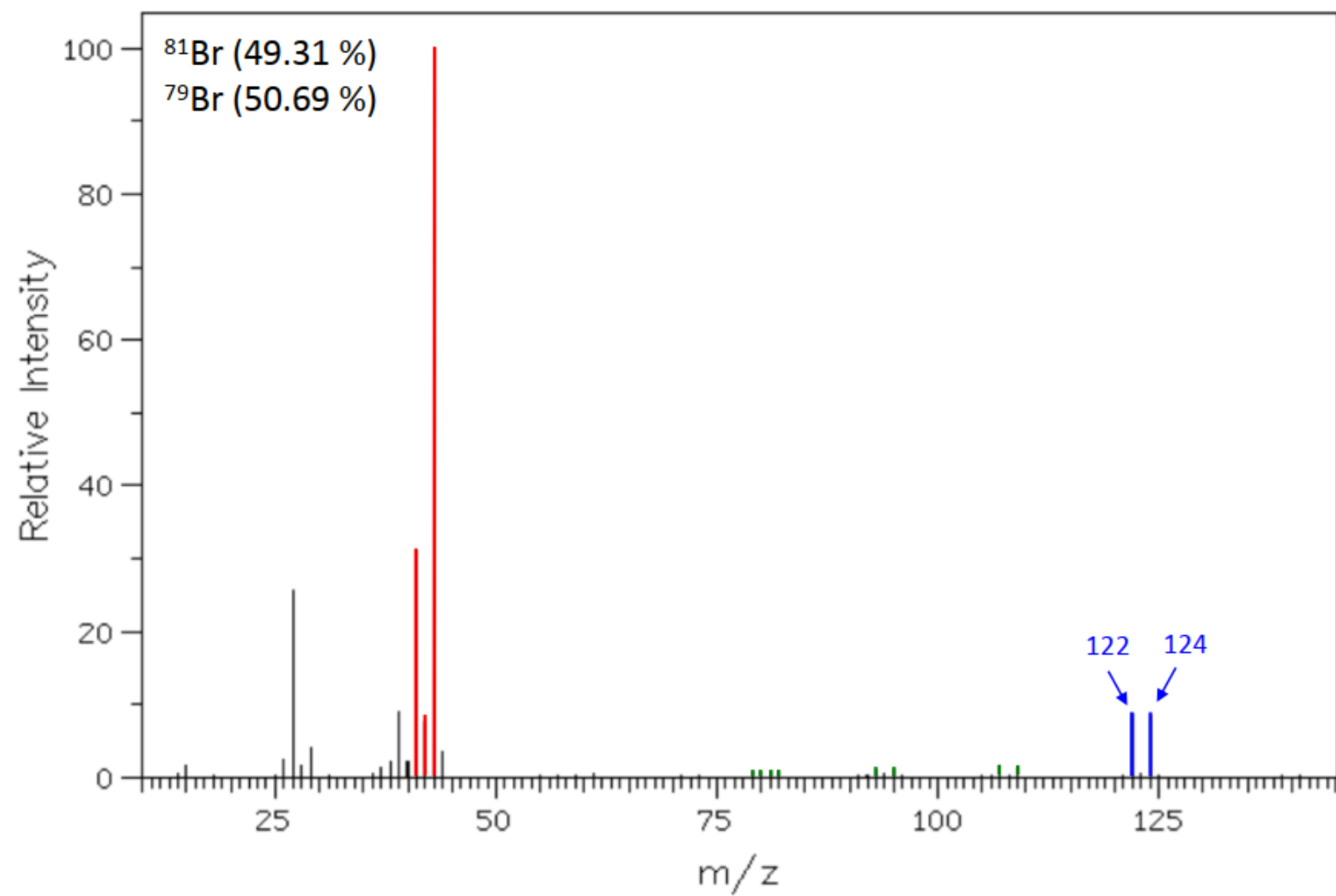


El mass spectrum

The mass spectrum of a chlorine-containing molecule is likewise easy to identify from two peaks separated by two mass units, but this time in a ratio of 3:1, arising from the 3:1 isotopic ratio of ^{35}Cl and ^{37}Cl .



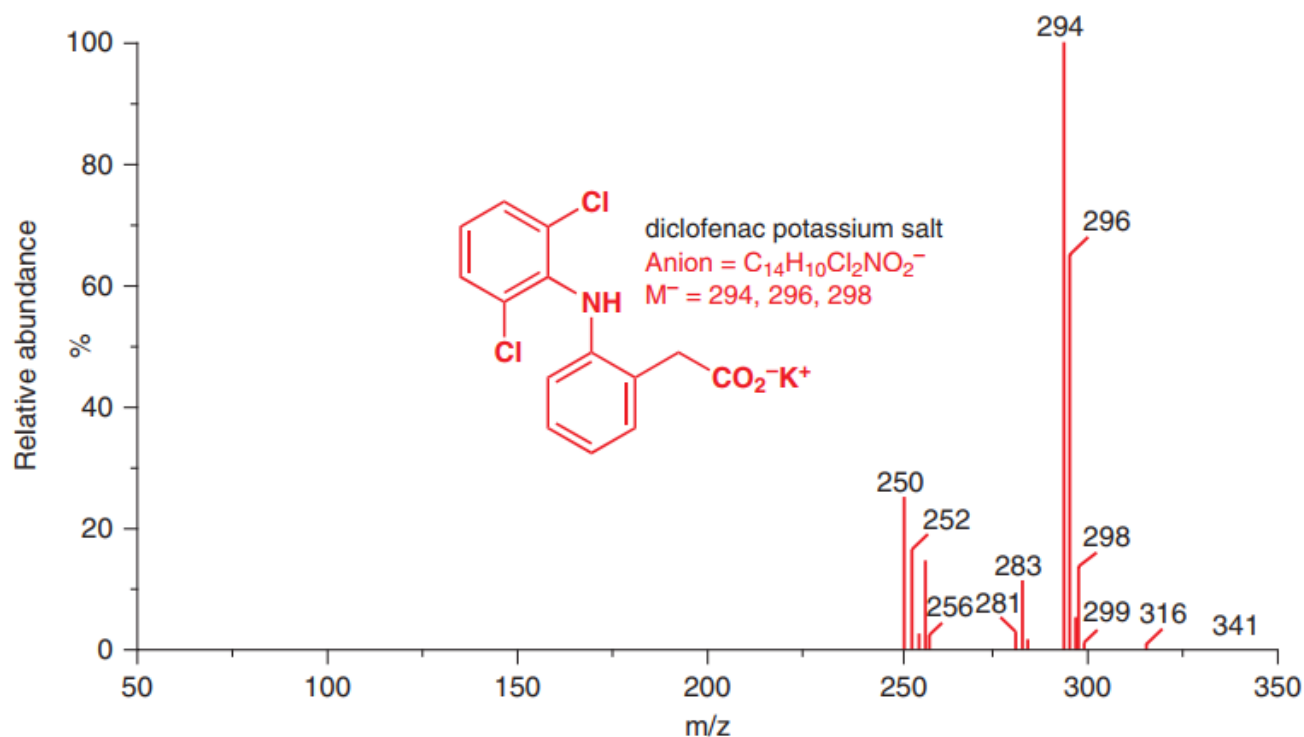
EI-Mass Spectrum of Isopropyl Chloride



EI-Mass Spectrum of Propyl Bromide

What happens with more than one Br or Cl?

The ES spectrum of diclofenac shows the mass of the carboxylate anion as three peaks, at 294, 296, and 298. The relative size of the peaks can be worked out from the 75% probability that each Cl atom will be ^{35}Cl and the 25% probability it will be ^{37}Cl . The ratios are therefore $3/4 \times 3/4$: $2 \times 3/4 \times 1/4$: $1/4 \times 1/4$ or 9 : 6 : 1.



Summary table of common elements with more than one isotope at >1% abundance

Element	Isotopes	Approximate ratio	Exact ratio
carbon	^{12}C , ^{13}C		98.9:1.1
chlorine	^{35}Cl , ^{37}Cl	3:1	75.8:24.2
bromine	^{79}Br , ^{81}Br	1:1	50.5:49.5

H, N, O, S, P, F, and I have only very small amounts of isotopes other than ^1H , ^{14}N , ^{16}O , ^{31}P , ^{32}S , and ^{127}I . The real oddity though is tin, which exists as a mixture of 10 different stable isotopes, the major ones being ^{116}Sn (15%), ^{117}Sn (8%), ^{118}Sn (24%), ^{119}Sn (9%), ^{120}Sn (33%), ^{122}Sn (5%), and ^{124}Sn (6%). In reality the precise ratio of isotopes for any element varies according to its source, a fact which can supply useful forensic information.

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Isotope	% nat. abundance	atomic mass
^1H	99.985	1.007825
^2H	0.015	2.0140
^{12}C	98.89	12 (formerly by definition)
^{13}C	1.11	13.00335
^{14}N	99.64	14.00307
^{15}N	0.36	15.00011
^{16}O	99.76	15.99491
^{17}O	0.04	16.99913
^{18}O	0.2	17.99916
^{28}Si	92.23	27.97693
^{29}Si	4.67	28.97649
^{30}Si	3.10	29.97376
^{32}S	95.0	31.97207
^{33}S	0.76	32.97146
^{34}S	4.22	33.96786
^{35}Cl	75.77	34.96885
^{37}Cl	24.23	36.96590
^{79}Br	50.69	78.9183
^{81}Br	49.31	80.9163

Isotopic Abundance and Masses for Selected Elements

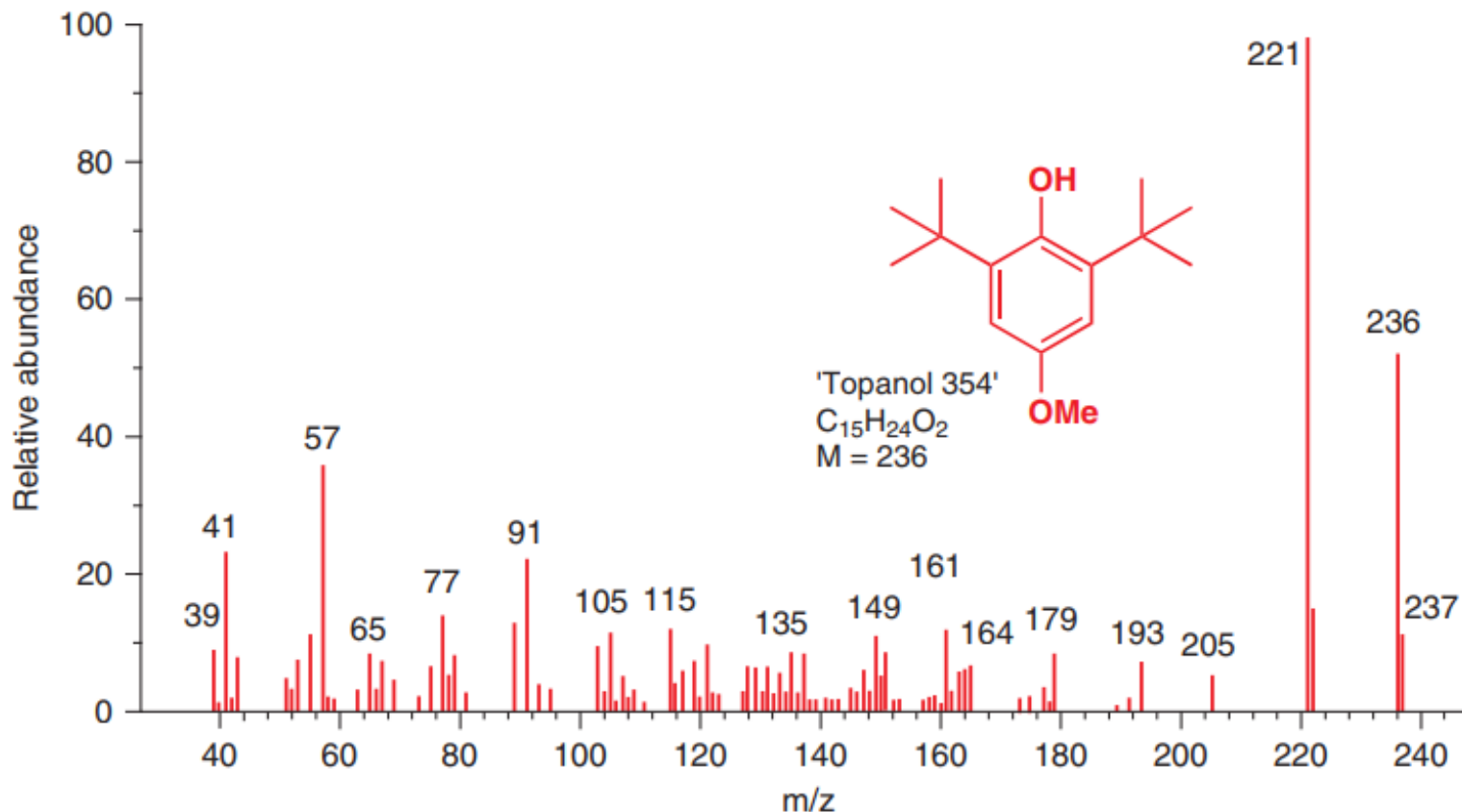
Element	Isotope	Exact Mass	Abundance %
Hydrogen	^1H	1.007825	99.985
	^2H or D	2.0140	0.015
Carbon	^{12}C	12.0000	98.90
	^{13}C	13.00335	1.10
Nitrogen	^{14}N	14.00307	99.63
	^{15}N	15.00011	0.37
Oxygen	^{16}O	15.99491	99.759
	^{17}O	16.99913	0.037
	^{18}O *	17.99916	0.204
Fluorine	^{19}F	18.99840	100.0
Silicon	^{28}Si	27.97693	92.21
	^{29}Si	28.97649	4.67
	^{30}Si	29.97377	3.10
Phosphorus	^{31}P	30.97376	100.0
Sulfur	^{32}S	31.97207	95.0
	^{33}S	32.97146	0.75
	^{34}S	33.96787	4.22
Chlorine	^{35}Cl	34.96885	75.77
	^{37}Cl	36.96590	24.23
Bromine	^{79}Br	78.91834	50.69
	^{81}Br	80.91629	49.31
Iodine	^{127}I	126.90447	100.0

Carbon has a minor but important isotope ^{13}C

The minor isotopes of many elements that appear at below the 1% level are not usually important, but one we cannot ignore is the 1.1% of ^{13}C present in ordinary carbon, of which the main isotope is of course ^{12}C .

The exact height of these peaks is useful as an indication of the number of carbon atoms in the molecule. Each carbon has a 1.1% chance of being ^{13}C rather than ^{12}C , so the more C atoms there are the bigger this chance becomes. If there are n carbon atoms in a molecular ion, then the ratio of M^+ to $[\text{M} + 1]^+$ is $100:(1.1 \times n)$.

Topanol 354: With 15 carbons, there's a 16.5% chance there will be one ^{13}C atom in the molecule, and you can clearly see the sizeable $\text{M} + 1$ peak at 237.



Atomic composition can be determined by high-resolution mass spectrometry (HRMS)

Bee pheromone had MW 114. When we revealed it was $\text{C}_7\text{H}_{14}\text{O}$, we had to use other information to infer this, because 114 could also be many other things, such as C_8H_{18} or $\text{C}_6\text{H}_{10}\text{O}_2$ or $\text{C}_6\text{H}_{14}\text{N}_2$. These different atomic compositions for the same molecular weight can nonetheless be distinguished if we know the *exact* molecular weight, since individual isotopes have non-integral masses (except ^{12}C by definition). The table below gives these masses to five decimal places, which is the sort of accuracy you need for meaningful results. Such accurate mass measurements are obtained by a technique called *high-resolution mass spectrometry*.

Exact masses of common elements

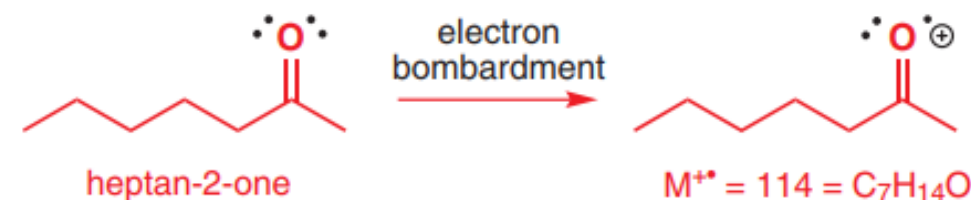
Element	Isotope	Mass number	Exact mass
hydrogen	^1H	1	1.00783
carbon	^{12}C	12	12.00000
carbon	^{13}C	13	13.00335
nitrogen	^{14}N	14	14.00307
oxygen	^{16}O	16	15.99492
fluorine	^{19}F	19	18.99840
phosphorus	^{31}P	31	30.97376
sulfur	^{32}S	32	31.97207
chlorine	^{35}Cl	35	34.96886
chlorine	^{37}Cl	37	36.96590
bromine	^{79}Br	79	78.91835
bromine	^{81}Br	81	80.91635

For the bee pheromone, the accurate mass turns out to be 114.1039. The table below compares possible atomic compositions for an approximate MW 114, and the result is conclusive. The exact masses to three places of decimals fit the observed exact mass only for the composition $C_7H_{14}O$. You may not think the fit is very good when you look at the two numbers, but notice the difference in the error expressed as parts per million. One answer stands out from the rest. Note that even two places of decimals would be enough to distinguish these four compositions.

One thing you may have noticed in the table above is that there are no entries with just one nitrogen atom. Two nitrogen atoms, yes; one nitrogen no! This is because any complete molecule with C, H, O, S, and *just one nitrogen in it has an odd molecular weight*. This is because C, O, S, and N all have even atomic weights—only H has an odd atomic weight. Nitrogen is the only element from C, O, S, and N that can form an odd number of bonds (3). Molecules with one nitrogen atom must have an odd number of hydrogen atoms and hence an odd molecular weight.

Exact mass determination for the bee alarm pheromone

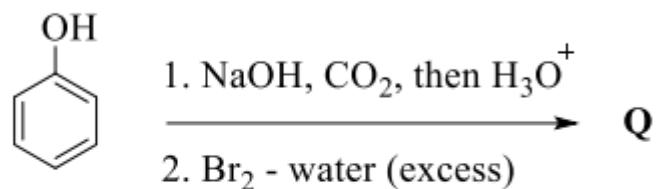
Composition	Calculated M^+	Observed M^+	Error in ppm
$C_6H_{10}O_2$	114.068075	114.1039	358
$C_6H_{14}N_2$	114.115693	114.1039	118
$C_7H_{14}O$	114.104457	114.1039	5
C_8H_{18}	114.140844	114.1039	369



Q.16 The major product in the given reaction sequence is **Q**. The mass spectrum of **Q** shows

Ans: B

([M] = molecular ion peak)



- (A) [M], [M+2], [M+4], and [M+6] peaks with relative intensity of 1:1:1:1
- (B) [M], [M+2], [M+4], and [M+6] peaks with relative intensity of 1:3:3:1
- (C) [M], [M+2], and [M+4] peaks with relative intensity of 1:2:1
- (D) [M] and [M+2] peaks with relative intensity of 1:1



Q.7	An organic compound exhibits the $[M]^+$, $[M+2]^+$ and $[M+4]^+$ peaks in the intensity ratio 1:2:1 in the mass spectrum, and shows a singlet at δ 7.49 in the ^1H NMR spectrum in CDCl_3 . The compound is:
(A)	1,4-dichlorobenzene
(B)	1,4-dibromobenzene
(C)	1,2-dibromobenzene
(D)	1,2-dichlorobenzene

CSIR-DECEMBER-2015

In the mass spectrum of 1,2-dichloroethane, approximate ratio of peaks at m/z values 98, 100, 102 will be

- | | |
|----------|----------|
| 1. 3:1:1 | 2. 9:6:1 |
| 3. 1:1:2 | 4. 1:2:1 |

What is the approximate ratio of the peaks at m/z values 98, 100, and 102 in the mass spectrum of 1,2-dichloroethane?

In the mass spectrum of 1,2-dichloroethane, approximate ratio of peaks at m/z values 98, 100, 102 will be---

Ans: 2

0.562: 0.375:0.0625

= 9:6:1

CSIR-JUNE-2017

Mass fragment of $[\text{IrCl}]^+$ in mass spectrometry shows three mass peaks at $m/z = 226, 228,$ and 230 . Given that natural abundances of ^{191}Ir , ^{193}Ir , ^{35}Cl , and ^{37}Cl are 37%, 63%, 76%, and 24% respectively, the intensities of the mass peaks are in the order

1. 49.5: 100: 26.6
2. 100: 49.5: 26.6
3. 26.6: 100: 49.5
4. 26.6: 49.5: 100

NET June-2018

Mass spectrum of a compound shows an $[M+2]$ ion peak that is 25% of M. This indicates that compound has one—

a) Fluorine b) Sulphur c) Bromine b) Chlorine

Silverstein P18

cleavage of the molecular ion. Because of the extremely low pressure in the mass spectrometer, there are very few fragment collisions; we are dealing largely with unimolecular decompositions. This assumption, backed by a large collection of reference spectra, is the basis for the vast amount of information available from the fragmentation pattern of a molecule. Whereas conventional organic chemistry deals with reactions initiated by chemical reagents, by thermal energy, or by light, mass spectrometry is concerned with the consequences suffered by an organic molecule at a vapor pressure of about 10^{-6} mm Hg struck by an ionizing electron beam.

A number of general rules for predicting prominent peaks in EI spectra can be written and rationalized by using standard concepts of physical organic chemistry.

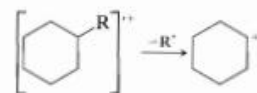
1. The relative height of the molecular ion peak is greatest for the straight-chain compound and decreases as the degree of branching increases (see rule 3).
2. The relative height of the molecular ion peak usually decreases with increasing molecular weight in a homologous series. Fatty esters appear to be an exception.
3. Cleavage is favored at alkyl-substituted carbon atoms: the more substituted, the more likely is cleavage. This is a consequence of the increased stability of a tertiary carbocation over a secondary, which in turn is more stable than a primary.

Cation stability order:



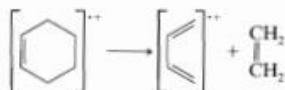
Generally, the largest substituent at a branch is eliminated most readily as a radical, presumably because a long-chain radical can achieve some stability by delocalization of the lone electron.

4. Double bonds, cyclic structures, and especially aromatic (or heteroaromatic) rings stabilize the molecular ion and thus increase the probability of its appearance.
5. Double bonds favor allylic cleavage and give the resonance-stabilized allylic carbocation. This rule does not hold for simple alkenes because of the ready migration of the double bond, but it does hold for cycloalkenes.
6. Saturated rings tend to lose alkyl side chains at the α bond. This is merely a special case of branching (rule 3). The positive charge tends to stay with the ring fragment. See Scheme 1.4.



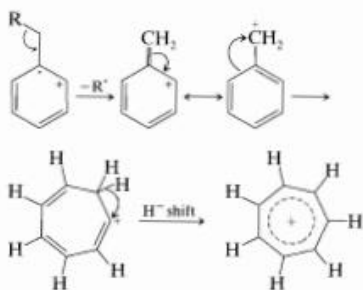
(Sch 1.4)

Unsaturated rings can undergo a *retro*-Diels-Alder reaction Scheme 1.5:



(Sch 1.5)

7. In alkyl-substituted aromatic compounds, cleavage is very probable at the bond β to the ring, giving the resonance-stabilized benzyl ion or, more likely, the tropylium ion (see Scheme 1.6).



(Sch 1.6)

8. The C—C bonds next to a heteroatom are frequently cleaved, leaving the charge on the fragment containing the heteroatom whose non-bonding electrons provide resonance stabilization.
9. Cleavage is often associated with elimination of small, stable, neutral molecules, such as carbon monoxide, olefins, water, ammonia, hydrogen sulfide, hydrogen cyanide, mercaptans, ketene, or alcohols, often with rearrangement (Section 1.5.5).

It should be kept in mind that the fragmentation rules above apply to EI mass spectrometry. Since other ionizing (CI, etc.) techniques often produce molecular ions with much lower energy or quasimolecular ions with very different fragmentation patterns, different rules govern the fragmentation of these molecular ions.

McLafferty Rearrangements

A very common fragmentation that can occur with many substrates is the **McLafferty rearrangement**. This fragmentation was first described by Fred McLafferty in 1956 and is one of the most predictable fragmentations, next to the simple α -cleavage. In the McLafferty rearrangement, a hydrogen atom on a carbon 3 atoms away from the radical cation of an alkene, arene, carbonyl, or imine (a so-called γ -hydrogen) is transferred to the charge site via a six-membered transition state, with concurrent cleavage of the sigma bond between the α and β positions of the tether.

This forms a new radical cation and an alkene with a pi bond between what were the original beta and gamma carbons. For simplicity, the mechanism of the McLafferty rearrangement is usually drawn as a concerted process, as in Figure. There is experimental evidence, however, that the fragmentation is in fact stepwise, and as a general rule fragmentations that involve breaking more than one bond are probably stepwise. The McLafferty rearrangement is readily observed in the mass spectra of many organic functional groups.

